

Swarm intelligence optimization-based fusion of ConvMixer-enhanced deep neural networks for brain tumor detection

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ABSTRACT

Accurate identification of brain tumors is crucial for early diagnosis and the development of effective treatment strategies. Deep learning models and ensemble techniques have been widely used for brain tumor detection, though many existing methods rely on simple ensemble approaches without utilizing advanced optimization techniques. This study proposes SwarmFusionDNN, a novel ensemble approach for brain tumor detection based on a weighted average ensemble. Our method incorporates ConvMixer block into popular pre-trained models, including MobileNet, DenseNet121, MobileNetV2, and Xception, to enhance feature extraction from brain MRI scans. The ConvMixer blocks improve the models ability to capture spatial and channel-wise information, leading to enhanced tumor detection performance. To enhance the ensemble's performance, a Particle Swarm Optimization (PSO) algorithm based on swarm intelligence is utilized to optimize the weights assigned to each model. The objective function for the optimization process is the minimization of classifier error rate, which ensures that the ensemble achieves the best possible performance. By assigning appropriate weights to each model, SwarmFusionDNN effectively combines their individual predictions, resulting in improved overall accuracy and robustness. The effectiveness of the proposed SwarmFusionDNN was evaluated through comprehensive experiments on three widely recognized benchmark MRI datasets: Br35H dataset (3000 images across 2 classes), Figshare dataset (3064 images across 3 classes), and the Brain Cancer MRI dataset from Mendeley (6056 images across 3 classes). The experiments show that SwarmFusionDNN achieved 99.66 % accuracy on the Br35H dataset, 96.14 % accuracy on the Figshare dataset, and 98.22 % accuracy on the Brain Cancer MRI dataset from Mendeley, demonstrating its strong performance across all three MRI datasets. Furthermore, Grad-CAM visualization confirms that the model focuses on tumor regions, providing additional insights into its robustness and accuracy. SwarmFusionDNN outperforms individual models, state-of-the-art optimization algorithms, and traditional ensemble methods in accuracy, demonstrating its effectiveness in brain tumor detection. By integrating ConvMixer blocks, weighted averaging, and swarm intelligence-based optimization, it shows significant potential to enhance medical diagnosis and treatment strategies for brain tumor detection.

1. Introduction

Brain tumors are complex and diverse neoplasms that present a significant challenge to the medical community. They can arise from various types of cells within the central nervous system and are classified into three main types: meningioma, glioma, and pituitary tumors [1]. The diagnosis of brain tumors is of utmost importance as it plays a

critical role in treatment planning and determining patient prognosis. In recent years, the integration of medical imaging techniques and artificial intelligence (AI) has revolutionized the field of medical diagnosis, particularly in the detection and characterization of brain tumors. Medical imaging has become an invaluable tool in diagnosing brain tumors and differentiating them from other intracranial abnormalities [2,3]. Methods like magnetic resonance imaging (MRI) offer

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comprehensive anatomical and functional data, allowing clinicians to visualize tumor presence, location, and extent with precision [4]. These imaging modalities aid in detecting abnormal growths, identifying their characteristics, and determining the appropriate treatment approach. The ability to non-invasively examine the brain through medical imaging has significantly improved the accuracy and efficiency of brain tumor diagnosis.

Advancements in AI have opened up new possibilities for the field of medical diagnosis, including the detection and classification of brain tumors. Deep learning (DL), a subset of AI, has demonstrated remarkable potential in the analysis and interpretation of medical images. DL algorithms utilize neural networks to process vast amounts of data and extract intricate patterns and features that may not be readily apparent to the human eye. By training on large datasets of annotated brain scans, DL algorithms can learn to identify subtle abnormalities indicative of brain tumors with high accuracy. This has the potential to reduce diagnostic errors and improve patient outcomes by enabling early detection and intervention [5]. Furthermore, transfer learning (TL) has further enhanced the capabilities of AI in medical diagnosis, including brain tumors. TL facilitates the application of knowledge acquired through training on one dataset to be transferred and utilized for a different yet related task. In the context of brain tumor diagnosis, TL enables the application of pre-trained models on large-scale datasets to smaller, specialized datasets. This approach overcomes the challenge of obtaining large annotated datasets specific to brain tumors, which can be time-consuming and resource-intensive. By leveraging the knowledge and expertise captured in pre-trained models, TL facilitates more accurate and efficient diagnosis of brain tumors [6]. Ensemble models have also emerged as a valuable tool in medical imaging for brain tumor diagnosis. Ensemble models combine the predictions of multiple AI models to make more robust and accurate diagnoses. By aggregating the results of different algorithms or incorporating diverse input data, ensemble models can overcome the limitations and biases of individual models, resulting in improved performance. This ensemble approach enhances the reliability and generalizability of AI-based brain tumor diagnosis, ultimately contributing to more effective treatment decisions [7].

While AI has shown great promise in the field of brain tumor diagnosis, it is important to note that these technologies should not replace human expertise but rather serve as valuable tools to assist medical professionals. Collaboration between AI systems and healthcare providers can lead to more precise and efficient diagnoses, enabling timely and personalized treatment strategies for patients with brain tumors. The integration of medical imaging techniques and AI algorithms, such as DL, TL, and ensemble models, has significantly advanced the diagnosis of brain tumors. These technological advancements have the potential to improve accuracy, efficiency, and patient outcomes in neuro-oncology. By harnessing the power of AI and medical imaging, clinicians can enhance their diagnostic capabilities, ultimately leading to more effective treatment interventions for individuals affected by brain tumors.

1.1. Motivation and major contributions

The timely detection of brain tumors is of paramount importance in their diagnosis, and MRI stands as a widely employed technique for this objective due to its exceptional image quality. However, the classification of brain tumors into multiple classes can be challenging due to their similarities. DL techniques have shown promise in accurately diagnosing brain tumors, often using TL to leverage existing models or introduce new modifications. TL is indeed beneficial when handling limited datasets; however, it also comes with some disadvantages. These include: Domain Mismatch, Model Compatibility, not all pre-trained models may be suitable for a specific task or dataset, making it crucial to choose the right base model. To address these challenges, hybrid techniques can indeed be employed, such as combining TL with other

methods like ensemble learning. Popular ensemble methods, including voting, stack [8] and weighted ensemble are frequently utilized to maximize performance and mitigate TL's limitations [9]. However, the determination of suitable weights in a weighted average ensemble for each model can be a pivotal factor in attaining optimal results. Conventional methods like experimental weight assignment techniques may demonstrate inefficiency and time-consuming tendencies [10,11]. In situations like this, a swarm intelligence-based approach can prove highly effective in determining the optimal weights for ensemble models [12]. This approach enhances efficiency and assists in the discovery of optimal weights. In this study, a novel architecture named 'SwarmFusionDNN' is introduced, employing Swarm Intelligence Optimization to fuse ConvMixer-enhanced Deep Neural Networks, thereby enhancing the accuracy of brain tumor detection.

In the past, research primarily concentrated on the creation of individual models for brain tumor detection. However, some studies have delved into the exploration of ensemble methods, like stacking and voting. However, as far as the authors are aware, this study marks the initial endeavor to introduce a swarm intelligence optimization-based ensemble model for brain tumor detection. In the proposed approach, four base models that utilize transfer learning are initially considered. This method leverages the knowledge learned from pre-trained models and applies it to the specific task. To enhance the base models, ConvMixer Blocks and classifier layers are incorporated. The ConvMixer Block is constructed from a series of depthwise separable convolutions, which are then followed by a pointwise convolution. This design enables ConvMixer Blocks to efficiently capture both spatial and channel-wise information, making them well-suited for feature extraction in our brain tumor detection task [13]. In the testing phase, outputs are generated from four enhanced base models. These outputs are integrated using a weighted average ensemble technique, where the challenge lies in determining optimal weights for each model to maximize predictive accuracy. This approach enhances efficiency and represents a novel contribution to brain tumor detection methods. To optimize these weights, a swarm intelligence optimization-based PSO algorithm is employed [14]. Drawing inspiration from the collective behavior observed in bird flocking or fish schooling, the PSO algorithm employs a population of particles to explore the solution space and discover the optimal weights for the ensemble. By leveraging swarm intelligence, the search for the best weight configuration can be conducted efficiently, leading to improved performance in brain tumor detection. The primary contributions of our research endeavor encompass the following:

1. SwarmFusionDNN is proposed as an ensemble approach for brain tumor detection. The architecture of the four base models is enhanced by incorporating ConvMixer blocks, which improves the models' capability to extract relevant features from MRI scans, resulting in more accurate tumor classification. The integration of ConvMixer blocks in each base model enhances their ability to capture intricate spatial and channel-wise information, ultimately boosting the overall performance of SwarmFusionDNN in brain tumor detection.
2. PSO algorithm based on swarm intelligence is employed to optimize the weighting of each base model within the SwarmFusionDNN. Through this optimization process, SwarmFusionDNN achieves noteworthy improvements in classification accuracy compared to random weight assignments. Comprehensive comparisons are conducted with other ensemble methods, including stacking, averaging, the Gompertz function, and Condorcet's jury theorem, to highlight the superior performance of SwarmFusionDNN in the context of brain tumor detection.
3. This study employs three brain MRI datasets to assess the effectiveness of the proposed approach. The results obtained provide compelling evidence of the superiority and practical applicability of SwarmFusionDNN compared to both single models and other ensemble techniques.

4. This study utilizes Grad-CAMs as a significant aspect of the research. By leveraging Grad-CAMs, tumors are effectively localized within brain MRI scans. Furthermore, extensive comparisons with previous approaches are conducted to highlight the superior performance of SwarmFusionDNN in brain tumor detection.

Section 2 presents a detailed literature review of existing brain tumor detection approaches and highlights the gap addressed by our proposed method. Section 3 discusses the materials and methods used in this study, including dataset details, model architecture, and optimization techniques. Section 4 presents the experimental results and discussion, showcasing the performance of SwarmFusionDNN on multiple datasets and comparing it to other approaches. Lastly, Section 5 concludes the paper with key findings and potential future directions.

2. Literature review

A variety of approaches have been suggested to address the task of classifying brain tumors using DL methodologies. The authors [15] introduced an ensemble-based model to detect brain tumors. In their study, they employed the inceptionV3 architecture for extracting relevant features, and employed machine learning algorithms to classify these features into relevant categories. Through their proposed ensemble model, they were able to achieve an accuracy rate of 94.34 %. The study [16] conducted a study where they employed three distinct pre-trained CNN models in combination with TL techniques to classify brain tumors. Among these models, VGG16 achieved the highest accuracy rate of 98.67 %. The authors [17] tried to expand the dataset and performed fine-tuning on the CNN method. The proposed method was applied to both the original dataset and the augmented dataset. As a result, the augmented dataset yielded an improved accuracy of 94.58 %. The authors [18] introduced a novel approach for the multi-classification of brain tumors, utilizing a dual-level ensemble model. This innovative ensemble-based method outperformed a single neural network, showcasing its superior performance in accurately classifying brain tumors. The authors [19] presented a novel CNN designed for the automatic segmentation of brain tumors. Through rigorous evaluation on an MRI dataset, the proposed model demonstrated remarkable effectiveness, achieving an accuracy of 97.3 %. The authors [8], they introduced a BMRI-NET for the detection of brain tumors from MR images. They present an ensemble approach that incorporates three pre-trained models, thereby enhancing the performance. Extensive experimentation clearly showcases the strength of the BMRI-NET model, as it achieves an accuracy of 98.69 %. This study [20] presented a novel framework for feature extraction based on VGG-19 architecture. Their experiment focused on an MRI dataset, and the results demonstrated an accuracy rate of 96.13 %. Furthermore, the authors [7] introduced a modified ResNet50 architecture specifically tailored for efficient classification of brain MRI images. Their approach involved incorporating an additional eight layers to the original ResNet50 model. The results of the study showcased success rate of 97 % for this strategy. In [21], the authors employed a combination of three distinct CNN models and constructed an ensemble model, resulting in a remarkable accuracy of 98.83 %. The authors [22] has presented a modified version of the ResNet50 model, integrating it with the Enhanced Watershed Segmentation algorithm for brain tumor classification. Through this innovative approach, they achieved a notable accuracy of 92 %. The authors [23] utilized a DL model to analyze a dataset consisting of 3064 MRI images. Remarkably, their approach yielded an accuracy level of 95.6 %. In [24], they introduced a CNN model comprising seven layers, augmented with the utilization of a genetic algorithm. Their proposed model showcased exceptional performance, achieving an accuracy of 94.2 % when evaluated on a brain MRI dataset. The authors [25] proposed an architecture by combining two CNN models, aiming to enhance the accuracy. Through their innovative approach, they achieved an accuracy rate of 97.37 %, which

stands as the highest reported accuracy in their study. The authors [26] introduced a novel Kronecker convolution method designed to eliminate redundancy within feature vectors and enhance detection accuracy. Through their proposed approach, they achieved an accuracy rate of 98.56 %. The study [27] designed a CNN model specifically designed to identify brain tumor. Their approach incorporated various enhancement techniques and utilized six distinct datasets for training and validation. Their proposed model achieved a remarkable classification accuracy of 98.8 % in detecting brain tumors. The authors [28], they employed the GoogleNet model to extract MRI features. Their methodology involved training and testing the proposed method using the Figshare dataset. Their approach yielded a notable accuracy of 97 %. Many recent advancements in healthcare, particularly in disease detection and classification, have leveraged swarm intelligence techniques. The authors [29] introduced the GSODTL-C3M model for colorectal cancer classification, combining Galactic Swarm Optimization with DL to improve accuracy. Similarly, Sohaib Asif et al. [30] developed a metaheuristics optimization-based ensemble model (MO-WAE) for detecting Mpox, achieving 97.78 % accuracy by optimizing model weights with PSO. These approaches highlight the power of swarm intelligence in enhancing medical diagnostics. Tanay Semwal et al. [31] proposed a hybrid model combining CNN and SVM, optimized using PSO, for brain tumor classification, achieving 84.77 % accuracy and outperforming traditional methods in detection. Despite the beneficial contributions made by various researchers in the domain of brain tumor classification using DL methods, there are still ample opportunities for further advancements. Ensemble learning approaches, which leverage the strengths of different classifiers, have been commonly employed in DL problems to enhance performance. However, assigning appropriate weights to the base classifiers is crucial for achieving optimal results [10]. To tackle this issue, a swarm intelligence-driven approach can be employed to search for the most optimal weights within the ensemble model [12,14]. This approach offers improved efficiency and facilitates the identification of optimal weights for the SwarmFusionDNN. This study introduces a novel architecture called SwarmFusionDNN, which incorporates Swarm Intelligence Optimization-based Fusion of ConvMixer-enhanced Deep Neural Networks. In this strategy, the weights allocated to each base model within the ensemble are optimized using the PSO algorithm based on swarm intelligence. Through this optimization process, SwarmFusionDNN achieves substantial enhancements in classification accuracy when contrasted with other ensemble methods and random weight assignments. This innovative approach demonstrates the potential of utilizing swarm intelligence optimization techniques to enhance the performance of ensemble models for brain tumor detection.

3. Materials and methods

This section presents a summary of the materials and methodologies utilized in our research. The dataset used consists of brain MRI images and underwent preprocessing steps to ensure data quality. Deep feature extraction was performed using pre-trained CNNs, while the ConvMixer block was incorporated to enhance feature representation. Additionally, the proposed SwarmFusionDNN architecture is introduced, which optimizes weights using the swarm intelligence-based PSO algorithm. Fig. 1 illustrates the overall methodology of our approach.

3.1. Dataset description

Experiments were conducted using three distinct brain MRI datasets, each providing valuable data for evaluating the effectiveness of the proposed method. The first two datasets, BR35H and Figshare, have been widely used for brain tumor classification tasks. A new dataset, the Bangladesh Brain Cancer MRI Dataset, was also utilized to further enhance the generalizability and robustness of the proposed model.

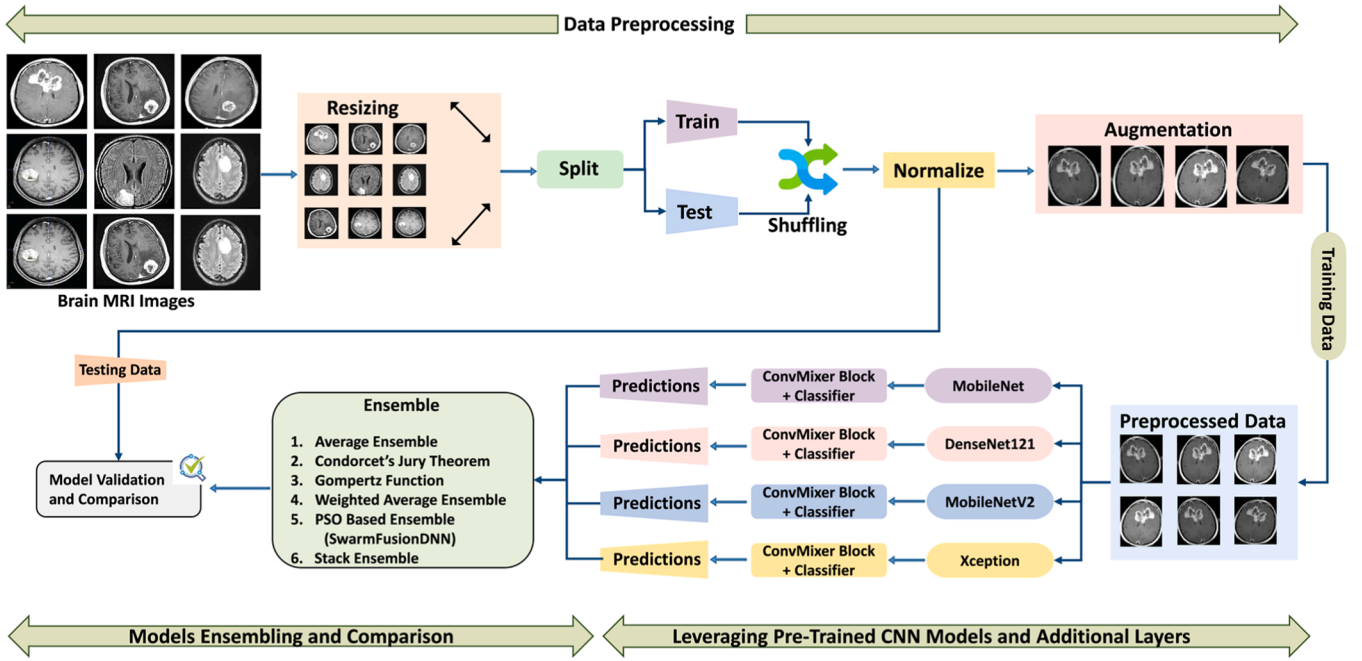


Fig. 1. Overview of the Proposed SwarmFusionDNN Architecture showcasing the step-by-step process for brain tumor detection.

3.1.1. BR35H dataset

The BR35H [32], obtained from the Kaggle repository, contains a total of 3000 MRI images: 1500 images of brain tumors and 1500 images of healthy (normal) brain tissue. This dataset is binary, consisting of two classes: tumor and normal. The images were collected from various sources but primarily consist of brain tumor types such as glioma and meningioma. The tumor sizes in the dataset vary, ranging from small, localized tumors to larger, more complex cases. The images in this dataset are typically obtained using standard T1-weighted MRI sequences.

3.1.2. Figshare dataset

The Figshare [33] comprises 3064 MRI slices from 233 patients, with three distinct types of brain tumors: glioma (1426 images), meningioma (708 images), and pituitary tumors (930 images). This dataset provides multi-class classification data, with images acquired from a range of MRI modalities, including T1-weighted, T2-weighted, and contrast-enhanced scans. The tumors vary in size and stage, with gliomas typically representing larger, more aggressive tumor types, while meningiomas and pituitary tumors tend to have more defined growth patterns. The dataset's diversity in terms of tumor type and MRI sequences contributes to its value in evaluating diagnostic models.

3.1.3. Bangladesh brain cancer MRI dataset

A new dataset, the Bangladesh Brain Cancer MRI Dataset [34], was also used in this study. This dataset consists of 6056 MRI images, categorized into three classes:

- Glioma: 2004 images
- Meningioma: 2004 images
- Tumor: 2048 images

These images were sourced from hospitals across Bangladesh and have been resized to 512×512 pixels. The dataset is notable for its diversity, containing various MRI sequences, imaging modalities, and patient demographics. As a result, this dataset is crucial for ensuring the robustness of the proposed method and its ability to generalize across different clinical scenarios. It is worth noting that the dataset includes variations in tumor size, from small to large tumors, and provides

insights into different tumor types.

By utilizing these datasets, which represent a diverse range of tumor types, MRI sequences, and patient demographics, we ensure that the proposed method is evaluated on a comprehensive and robust set of data. This approach allows for an effective and generalized evaluation of the model's performance in a manner comparable to the latest brain tumor diagnostic models. Fig. 2 illustrates representative samples from the datasets, providing visual examples of the brain MRI images used in our study. Table 1 summarizes the key details of the three brain MRI datasets used in our experiments.

3.2. Pre-processing and augmentation

Preprocessing and augmentation play crucial roles in enhancing the effectiveness of DL models for medical image analysis. Preprocessing techniques are employed to transform raw input data into a format suitable for analysis, while augmentation techniques help increase the diversity and robustness of the training data. In this study, all images were resized to a standardized dimension of 224×224 pixels, ensuring consistency across the dataset. Augmentation techniques, including flipping, shear transformations, and rotation, were applied to enhance the variability of the training data. These techniques improve model generalization and robustness by simulating real-world variations. Specifically, horizontal and vertical flips, shear range (0.2), and rotation ($\pm 20^\circ$) were employed to expose the model to different orientations, deformations, and spatial shifts, ultimately improving its ability to handle diverse MRI scan conditions. This augmentation process aids in capturing a wider range of image variations, ultimately improving the model's ability to generalize and make accurate predictions on unseen data. To ensure effective model training and reliable evaluation, the dataset was partitioned using the widely adopted 80–20 split ratio, where 80 % of the data was allocated for training purposes and the remaining 20 % for testing. This common split ratio is widely adopted in DL research, including similar studies on medical image analysis, as it allows for effective model training and evaluation while preserving a sufficient amount of data for validation purposes [21,35].

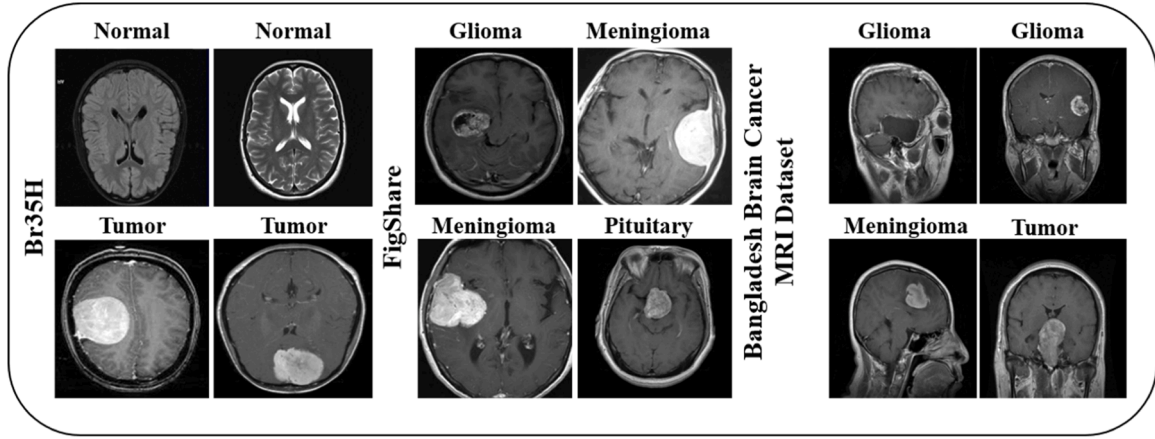


Fig. 2. Representative brain MRI images from the dataset, showcasing different tumor types used for evaluation in our study.

Table 1
Details of MRI datasets.

Dataset	Source	Total Images	Tumor Types
BR35H	Kaggle	3000	Tumor: 1500, Normal: 1500
Figshare	Figshare	3064	Glioma: 1426, Meningioma: 708, Pituitary Tumor: 930
Brain Cancer - MRI	Mendeley	6056	Glioma: 2004, Meningioma: 2004, Tumor: 2048

3.3. Deep feature extraction using pre-trained CNNs

Deep feature extraction plays a crucial role in capturing essential and distinctive information from raw images and representing it in a lower-dimensional space. In this study, pre-trained networks, namely MobileNet, DenseNet121, MobileNetV2, and Xception, were leveraged to extract high-level deep features from the MRI images. These architectures were adapted and fine-tuned specifically for the MRI datasets. As illustrated in Fig. 3, after extracting the deep features, a ConvMixer block was introduced [13]. The ConvMixer block, integrated into the prediction head, aims to explore the spatial and channel relationships within the extracted features. This exploration is accomplished through the utilization of depthwise and pointwise convolutions in the ConvMixer block. By incorporating Convolution Mixer, redundancy in both spatial and channel features is minimized, while simultaneously increasing the receptive field. Ultimately, a classifier is applied to classify these refined features into their respective classes. Below are specific details about each architecture utilized in this study:

MobileNet: MobileNet [36] is a lightweight CNN architecture known for its efficient computation on mobile devices. It utilizes depthwise separable convolutions to reduce computational complexity

while maintaining accuracy, making it suitable for resource-constrained environments.

DenseNet121: DenseNet [37] is a densely connected CNN architecture that promotes feature reuse and information flow throughout the network. DenseNet121, with its 121 layers, consists of densely connected convolutional blocks and transition layers, enabling effective feature extraction and learning.

MobileNetV2: MobileNetV2 [38] is an improved version of MobileNet, incorporating inverted residual blocks and linear bottlenecks. It combines depthwise separable convolutions with skip connections to enhance performance and efficiency, making it ideal for mobile and embedded applications.

Xception: Xception [39] is a CNN architecture based on the inception module, which employs depthwise separable convolutions to capture complex patterns and reduce parameter count. With its deep network structure, Xception enables high-level feature extraction and effective representation learning.

For each architecture, the pre-trained models were fine-tuned on the MRI datasets, adapting them to the specific characteristics of brain tumor detection tasks. These architectures served as powerful feature extractors, capturing meaningful representations from the MRI images, which were further refined using the ConvMixer block to enhance the spatial and channel relationships.

3.3.1. Rationale for model selection

Table 2 presents the performance and computational characteristics of six CNN architectures tested for brain tumor classification. The four models selected for the ensemble—MobileNet, DenseNet121, MobileNetV2, and Xception—were chosen based on their balanced trade-offs between accuracy, computational efficiency, and model complexity. MobileNet achieved the highest accuracy (91.32 %) with a relatively small model size (14.3 MB) and low parameter count (3.23 M), making it an efficient choice for deployment in resource-constrained

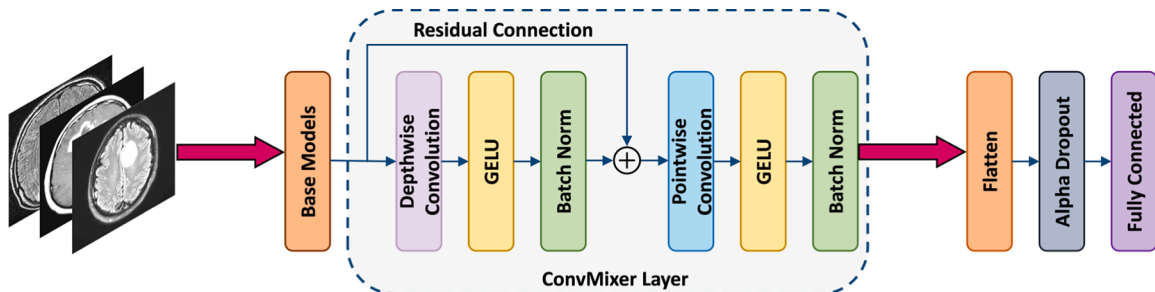


Fig. 3. Modified Architecture Overview illustrating the customized modifications made to pre-trained models for brain tumor detection, including the incorporation of the ConvMixer block to enhance feature extraction and classification performance.

Table 2

Performance and computational characteristics of the models tested for brain tumor classification.

Models	Accuracy	Parameters (M)	Model Size (MB)
MobileNet	91.32 %	3.23	14.3
DenseNet121	87.90 %	7.04	29.7
MobileNetV2	84.10 %	2.26	11.2
Xception	81.70 %	20.86	83.4
InceptionV3	78.77 %	21.80	84.10
EfficientNetB0	79.90 %	4.05	16.49

environments such as clinical applications. DenseNet121, while slightly larger, provided a competitive accuracy of 87.90 % with its deep architecture, benefiting from feature reuse through dense connections. This characteristic enhances its ability to extract complex features from brain MRI scans. MobileNetV2 offered a good balance of efficiency and performance, achieving 84.10 % accuracy while maintaining the smallest model size (11.2 MB) and the lowest parameter count (2.26 M), which makes it a lightweight option suitable for quick inference. Xception was included due to its depthwise separable convolutional architecture, which effectively captures spatial and channel-wise feature interactions, achieving an accuracy of 81.70 %. Although Xception is larger than the other models, its design makes it well-suited for extracting intricate features from medical images. Models such as InceptionV3 and EfficientNetB0 were not selected for the ensemble. InceptionV3, with an accuracy of 78.79 %, and EfficientNetB0, achieving 79.90 %, did not outperform the selected models in terms of accuracy despite having larger model sizes and parameter counts.

3.4. ConvMixer block

The ConvMixer block [13] is a novel architectural component designed to capture spatial and channel relationships within the feature maps. It employs depthwise and pointwise convolutions to exploit local and global information effectively. By leveraging depthwise convolutions, the block can capture spatial correlations efficiently, while the subsequent pointwise convolutions enable modeling of channel interactions. This combination enables the ConvMixer block to extract rich and meaningful feature representations. One of the significant advantages of the ConvMixer block is its ability to reduce redundancy in spatial and channel dimensions, leading to more efficient and compact feature representations. This reduction in redundancy helps alleviate overfitting and enhances the generalization capabilities of the models.

The integration of the ConvMixer block into the pre-trained models brings additional expressive power, enabling the models to capture finer spatial details and channel dependencies [13,40]. By enhancing the feature extraction process, the ConvMixer block aids in capturing more meaningful and discriminative information [41]. This ultimately leads to improved performance in tasks such as brain tumor detection, as it enables the models to learn more representative and informative features. By incorporating the ConvMixer block, along with the GeLU activation function and batch normalization, the efficiency and effectiveness of the feature extraction process are further enhanced. GeLU provides non-linearity, promoting better gradient flow and enhanced learning, while batch normalization stabilizes the network training by normalizing feature activations. Overall, the integration of the ConvMixer block into the pre-trained models enhances the deep feature extraction process, enabling the models to capture spatial and channel relationships effectively. The integration of this approach plays a vital role in enhancing the performance and accuracy of models in tasks such as brain tumor detection.

3.5. Leveraging pre-trained CNN models and customized layers

To extract deep features from resized 224×224 MRI images, we

utilize pre-trained models, including MobileNet, DenseNet121, MobileNetV2, and Xception, as feature extractors. After extracting features from the pre-trained models, ConvMixer blocks are integrated to further enhance the feature extraction process. The ConvMixer block is designed with simplicity in mind, consisting of basic convolution layers followed by batch normalization. A key aspect of the ConvMixer block is its ability to maintain the input structure throughout the layers. This feature makes it highly compatible with pre-trained architectures, allowing it to be seamlessly integrated without modifying the base model's architecture. By preserving the input structure, the ConvMixer block effectively captures both spatial and channel relationships within the feature maps. Furthermore, the ConvMixer block addresses redundancy by employing depthwise and pointwise convolutions, which efficiently model spatial correlations and channel interactions. This design results in more compact, informative feature representations that enhance the model's ability to differentiate between brain tumor and normal tissue. This approach ensures that ConvMixer modules effectively complement the pre-trained models, enhancing their feature extraction capabilities and improving overall classification performance.

To prevent overfitting and enhance generalization, alpha dropout with a rate of 0.5 is applied. Alpha dropout randomly masks inputs during training, forcing the model to learn more robust and independent features. This regularization technique helps to prevent co-adaptation of features and improve the model's ability to generalize to unseen data [42]. Finally, a dense layer is utilized for classification, mapping the extracted features to the respective brain tumor classes. The integration of the ConvMixer block and alpha dropout within the architecture enriches the feature extraction process, enabling the model to capture finer details and improve classification accuracy.

In summary, the incorporation of the ConvMixer block and alpha dropout enhances the feature extraction capabilities of the pre-trained models. This leads to improved performance in brain tumor classification by leveraging spatial and channel relationships, reducing redundancy, and promoting better generalization through regularization.

3.6. Proposed SwarmFusionDNN architecture

Training DL models can pose challenges because of data noise and the inherent algorithmic randomness, resulting in elevated variance [43] and potential generalization errors. While techniques like data augmentation and regularization help, more advanced methods are necessary. Ensemble learning combines the strengths of DL and ensemble techniques to enhance generalization. By creating an ensemble architecture, our goal is to accurately detect and classify brain tumor patients while outperforming existing methods. This approach aims to improve performance, reduce errors, and mitigate overfitting, ultimately leading to more accurate and robust brain tumor detection and classification. To boost the effectiveness of our SwarmFusionDNN, weighted average ensemble technique is integrated that amalgamates thoughtfully selected and customized base models. This ensemble technique, enhanced by the swarm intelligence PSO, is illustrated in Fig. 4 and acts as a potent fusion mechanism, leading to a considerable improvement in the overall performance of our model [44]. This innovative approach is a key highlight of our research. In previous research, a common practice involved employing a straightforward weighted ensemble method, necessitating multiple experiments to identify the optimal weights for base models in the ensemble. However, this approach is time-consuming and can yield unreliable outcomes. Consequently, determining the ideal weights for base models within the ensemble model presents a laborious and intricate challenge [45]. There are scenarios where highly competent models should contribute more to the ensemble prediction, while in other cases, less competent models should contribute less. Penalizing less accurate models and rewarding more accurate models can lead to further improvements in prediction accuracy.

To tackle this challenge, some researchers have utilized various

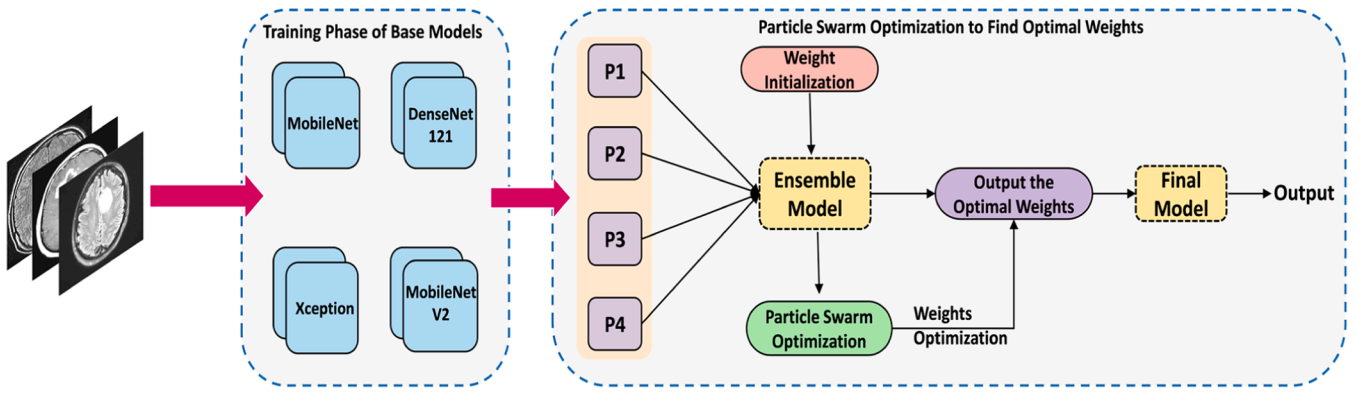


Fig. 4. Proposed ensemble architecture for brain tumor detection and classification, utilizing the weighted average ensemble approach with optimized model weights using the PSO algorithm.

optimization techniques, including genetic algorithms, to identify the best-weighted combinations for the base models employed in the ensemble process [46]. In our research, swarm intelligence optimization algorithm known as PSO is employed to automatically determine the optimized weights for each base model incorporated into our SwarmFusionDNN. This approach allows to achieve exceptional performance without the need for manual weight assignments to individual base models. The PSO algorithm [47] is chosen due to its faster convergence, higher search space density, and ability to reach good solutions without requiring local search. Our objective function in this optimization process is the reduction of classification error rates, as it aligns with our goal of achieving accurate brain tumor detection and classification. Through the utilization of the PSO algorithm, the most suitable weights for each base model within the SwarmFusionDNN architecture can be proficiently ascertained. This strategy harnesses the individual strengths of these models and generates a synergistic effect, ultimately resulting in a notable enhancement in overall performance for brain tumor detection. To provide clarity on the steps involved in the proposed ensemble architecture, Algorithm 1 is presented, outlining the main procedures and processes implemented in this approach.

3.6.1. PSO: A swarm intelligence approach for weight optimization

Optimization is a systematic process that aims to find the best solution by utilizing available information rather than random or exhaustive sampling. In this research, the focus is on optimizing the weights of each base model, which is crucial for achieving better performance. To accomplish this, the PSO algorithm is adopted, drawing inspiration from the collective behavior of bird flocks or fish. The PSO algorithm has been widely successful in various domains, including model training and optimization tasks. It leverages the concept of a swarm, where particles iteratively explore the search space to find optimal solutions. In our specific case, the PSO algorithm to optimize the weighting of each base model. The algorithm iterates for a specified number of iterations, typically can be set to 200, and in each iteration, it evaluates the error rate and updates the weights accordingly. In this research, the objective function for the PSO algorithm is defined as the reduction of the classification error rates. The classification error rate measures the misclassification of samples, and minimizing this error is essential for

improving the model's accuracy. Consequently, the objective is to reduce the classification error rate or, equivalently, maximize the model's accuracy. By optimizing the weights through the PSO algorithm, the goal is to enhance the model's ability to accurately classify brain tumor images. In choosing PSO over other optimization algorithms—such as genetic algorithms (GA), simulated annealing, or gradient-based optimizers—several key factors were considered. First, PSO is particularly well-suited for continuous, multidimensional optimization problems like weight assignment in ensemble learning, where no gradient information is available. Unlike gradient-based techniques, PSO does not require a differentiable objective function and is therefore more flexible in dealing with non-convex, non-linear loss landscapes. Moreover, compared to GA, PSO requires fewer parameters, is simpler to implement, and typically converges faster. Its population-based approach ensures a global search ability, reducing the risk of getting trapped in local minima. These characteristics make PSO a particularly robust and efficient method for optimizing ensemble weights in a high-dimensional classification task such as brain tumor detection using SwarmFusionDNN.

The objective function serves as a guiding metric that helps the optimization process find the best solution by quantifying the desired outcome. This research aims to find a set of weights that optimize the performance of SwarmFusionDNN for brain tumor detection in MRI images. The objective function is defined as the classifier error rate [48], which measures the misclassification rate of the ensemble model with the current set of weights. The rationale behind adopting the classifier error rate as the objective function in our brain tumor detection research is supported by its widespread utilization in various diagnostic applications [29].

The classifier error rate is calculated by dividing the number of misclassified samples by the total sample count, and then multiplying the result by 100 to express it as a percentage. Let's denote the objective function as *ClassifierErrorRate*, which aims to minimize the error rate. *ClassifierErrorRate* is designed to work by adjusting the weights assigned to each model within the ensemble, striving to find an optimal configuration that minimizes the overall classifier error rate, thus enhancing the accuracy and performance of the model. The objective function can be formulated as:

Algorithm 1

SwarmFusionDNN: A PSO-based Weighted Average Ensemble for MRI Classification.

- 1: Load the MRI dataset. 2: Perform preprocessing and augmentation techniques on the dataset (resizing, flipping, shear range, rotation, etc.). 3: Split the dataset into training and testing sets. 4: Extract deep features from pre-trained models (MobileNet, DenseNet121, MobileNetV2, Xception). 5: Modify each pre-trained model by adding the ConvMixer block and classifier head. 6: Train each modified model on the training dataset and save the model weights. 7: Perform predictions on the testing dataset using each trained model. 8: Combine the predictions of all modified base models using the weighted average approach. 9: Utilize the swarm intelligence algorithm for optimizing the base model weights, with a focus on minimizing the classifier error rate as the objective function. 10: Save the final optimized ensemble model. 11: Perform predictions on the testing dataset using the optimized ensemble model. 12: Compare the performance of SwarmFusionDNN with other pre-trained base models, ensemble methods, and existing methods:

- Evaluate the accuracy and performance metrics.

$$\text{ClassifierErrorRate} = \frac{\text{Missclassified Samples}}{\text{Total Samples}} \times 100 \quad (1)$$

Eq. (1) calculates the error rate as a percentage. The goal of the PSO algorithm is to minimize this error rate, which translates to maximizing classification accuracy. The algorithm assigns higher weights to more accurate models and lower weights to less accurate models, effectively leveraging the strengths of individual models.

The particle properties are represented by both position and velocity. These properties are denoted as $\text{aspr}_i = (pr_{i,1} + pr_{i,2} \dots \dots pr_{i,D})$ and $\text{ve}_i = (ve_{i,1} + ve_{i,2} \dots \dots ve_{i,D})$, where D shows the dimensionality. The algorithm's operation includes modifying the velocity and position of each particle, a process guided by the mathematical expressions presented below:

$$\text{ve}_{i,D}^{\text{itr}+1} = w \cdot \text{ve}_{i,D}^{\text{itr}} + c_1 r_1 (pr_{i,D} - pr_{i,D}^{\text{itr}}) + c_2 r_2 (g_D - pr_{i,D}^{\text{itr}}) \quad (2)$$

$$pr_{i,D}^{\text{itr}+1} = pr_{i,D}^{\text{itr}} + \text{ve}_{i,D}^{\text{itr}+1} \quad (3)$$

Here, $\text{ve}_{i,D}^{\text{itr}}$ and $pr_{i,D}^{\text{itr}}$ signify the velocity and position in D at iteration itr , w denotes the inertia weight, c_1 and c_2 stand for acceleration coefficients, r_1 and r_2 are random values distributed between 0 and 1.

Eq. (2) updates the velocity of each particle based on its current position, the best position it has encountered, and the global best position. The velocity determines how much the particle's position (the weight) changes during each iteration. This velocity update helps particles explore the search space to find optimal weights. Eq. (3) updates the position of each particle. The position corresponds to the weights of the models in the ensemble, and the update ensures that the weight values are adjusted based on the velocity.

During the PSO optimization process, the weights assigned to each base model are constrained within a specific range to ensure meaningful and stable model behavior. Specifically, the weights are restricted to the interval $[0, 1]$, meaning no weight can fall below 0 or exceed 1. This constraint prevents any individual model from dominating the ensemble and ensures a balanced contribution from all base models. These constraints are applied at each iteration of the optimization process, ensuring that the weight updates respect this boundary. The lower bound (lb) is set to 0, and the upper bound (ub) is set to 1. This restriction ensures that the optimization remains feasible and stable, avoiding extreme weight values that might negatively affect the ensemble's performance.

The PSO algorithm runs for a specified number of iterations, during which the particles explore the search space to find the optimal set of weights that minimizes the classifier error rate. After the iterations, the weights of the best particle or global best g are selected as the optimized weights for the ensemble model.

To derive the weighted ensemble prediction, the algorithm computes a weighted average of the predictions made by each individual model, utilizing the optimized weight values. Let's denote the ensemble prediction as y_{pred} and the individual model predictions as p_i . The final ensemble prediction equation is given by:

$$Y_{\text{pred}} = \frac{\sum_{i=1}^n w_i \cdot p_i}{\sum_{i=1}^n w_i} \quad (4)$$

Eq. (4) computes the weighted ensemble prediction by taking the weighted average of the predictions from each model. The optimized weights w_i from the PSO algorithm are applied to each model's prediction, ensuring that more accurate models have a greater influence on the final prediction.

An exhaustive analysis has been performed to identify the optimal parameter values for the PSO algorithm. A grid search approach was employed to explore a range of possible values for both the Population Size and Iterations, with the goal of determining the combination that maximizes performance. The following ranges were considered during

the tuning process:

- Population Size: Tested from 50 to 200.
- Iterations: Tested from 100 to 300.

The results of this parameter tuning process are summarized in Table 3, which includes the final selected values for Population Size and Iterations. These values were chosen based on their ability to optimize the performance of the PSO algorithm, ensuring efficient convergence while maintaining robustness.

Fig. 5 presents the convergence curve of the PSO-based ensemble model, showing the relationship between the classifier error rate and the number of iterations. Initially, the classifier error rate is relatively high, with significant reductions occurring in the early iterations, particularly between 0 and 20 iterations. After this rapid decline, the error rate stabilizes, demonstrating that the model reaches a convergence point. This behavior highlights the effectiveness of the PSO algorithm in minimizing classifier error, with fast early-stage learning followed by more refined optimization in the later stages, ultimately driving the model towards its optimal performance. By optimizing the weights using PSO, the algorithm guides the ensemble to assign higher weights to more accurate models and lower weights to less accurate models. This allows the ensemble to leverage the strengths of individual models and improve its overall prediction performance. The objective function, defined as the classifier error rate, quantifies the rate of misclassifications and steers the optimization process toward discovering the optimal set of weights that maximize the performance of SwarmFusionDNN. This approach ensures that the ensemble combines the predictions of each model in a way that maximizes its overall effectiveness in detecting brain tumors. Fig. 6 visually depicts the iterative nature of the PSO algorithm, where particles move through the search space in search of the optimal solution. The algorithm progressively refines the weights of the ensemble models to minimize the classifier error rate and enhance the ensemble's performance.

4. Experimental results and discussion

This section provides an overview of the evaluation protocols, experimental setup, and subsequent analysis. The dataset was carefully divided into training and testing sets, and data augmentation techniques were applied to improve model generalization. The base models were trained using TL and fine-tuning techniques. Performance evaluation was conducted on Dataset 1 and Dataset 2, measuring accuracy, precision, recall, and F1-score. Grad-CAM visualization was employed to understand the models' decision-making processes. The ensemble approach was compared with other popular ensemble methods, and the methodology was also contrasted with existing techniques. These results offer a comprehensive evaluation of the proposed methodology's effectiveness in brain tumor detection.

4.1. Evaluation protocols and experimental setup

In the Evaluation Protocols and Experimental Setup, various evaluation metrics are employed to assess the performance of our models. These metrics included the confusion matrix, ROC curve comparison,

Table 3
Summary of PSO Algorithm Parameters.

Parameters	Values
Population Size	50
C1	1
C2	2
Iterations	100
w	0.5
Variable Bounds	lb: 0, ub: 1

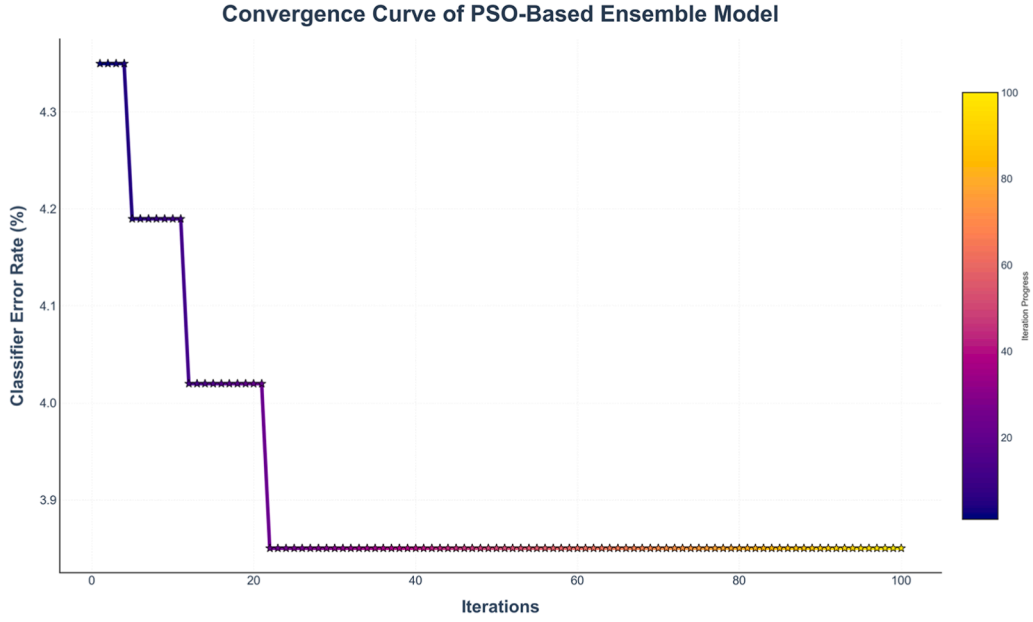


Fig. 5. Convergence curve of the PSO-based ensemble model.

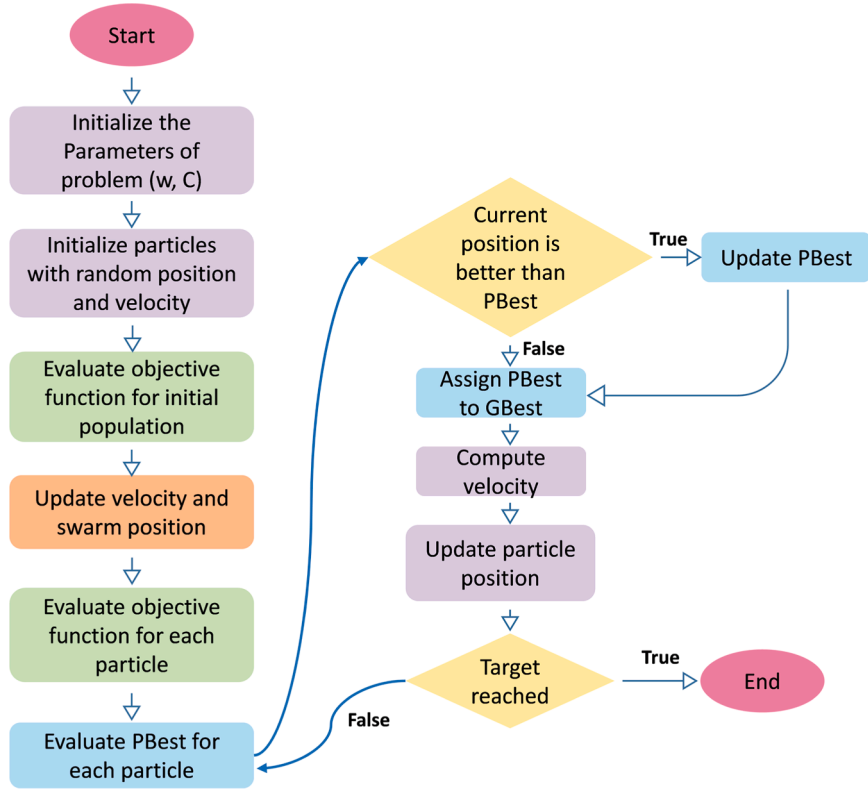


Fig. 6. PSO Optimization Process - Iterative movement of particles in search of optimal weights for the ensemble models.

and other evaluation measures defined in Eqs. (5)–9. These parameters rely on the outcomes derived from the confusion matrix, which includes True Positive (T_{Pst}), True Negative (T_{ngt}), False Positive (F_{Pst}), and False Negative (F_{ngt}). Among these, sensitivity reflects the model's ability to correctly identify positive cases, which is crucial for detecting tumors. Specificity, on the other hand, specificity measures the ability of the model to correctly identify negative cases, helping to reduce false alarms.

$$Accuracy (ACCU) = \frac{T_{Pst} + T_{ngt}}{T_{Pst} + T_{ngt} + F_{Pst} + F_{ngt}} \quad (5)$$

$$Sensitivity (SENS) = \frac{T_{Pst}}{T_{Pst} + F_{ngt}} \quad (6)$$

$$Precision (PREC) = \frac{T_{Pst}}{T_{Pst} + F_{Pst}} \quad (7)$$

$$F1 - \text{Score (F1)} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

$$\text{Specificity (SPEC)} = \frac{T_{ngt}}{T_{ngt} + F_{pst}} \quad (9)$$

For the experimental setup, TensorFlow and Keras frameworks are utilized to implement and train our models. The training process was conducted on a system equipped with a Ryzen 5 processor, Nvidia MX450 GPU, and 16 GB of RAM. These hardware specifications ensured efficient model training and processing capabilities.

4.2. Models training and implementation

In the Models Training and Implementation phase, careful selection of hyperparameters played a crucial role in achieving optimal model performance [49]. The Adam optimizer was employed, which offers advantages such as adaptive learning rates and momentum [50]. The cross-entropy loss function was utilized as it is well-suited for classification tasks. The training phase involved 20 epochs, using a learning rate of 1e-3, and employing a batch size of 64. To enhance training efficiency and prevent overfitting, the ReduceLROnPlateau technique was employed, which dynamically adjusted the learning rate by a factor of 0.5 when the validation accuracy plateaued. Additionally, the training process was terminated early if no further improvements in performance were detected, using the EarlyStopping callback with a patience value of 12 epochs. This ensured that training stopped when the model's performance plateaued, preventing unnecessary computations and overfitting. For transparency and reproducibility, the official code and implementation details of the proposed SwarmFusionDNN framework will be made available at the following GitHub repository: <https://github.com/sohaibasif21/SwarmFusionDNN>.

Fig. 7 presents the training and validation accuracy/loss curves for four DL models—MobileNet, DenseNet121, MobileNetV2, and Xception—trained on the Br35H dataset. All models exhibit a consistent upward trend in accuracy and a declining trend in loss across 20 epochs. MobileNet demonstrates the highest training and validation accuracy (~99 %) and the lowest loss, followed by DenseNet121 and Xception. MobileNetV2 shows slightly lower performance but maintains a stable convergence pattern. The close alignment of training and validation curves across all models indicates strong generalization capability and minimal overfitting. For this study, we focused on the Br35H dataset to provide a clear representation of model performance. While similar trends were observed on the other datasets, we chose to present results for one dataset to maintain focus and avoid excessive redundancy.

4.3. Performance evaluation on Br35H dataset

This section assesses the effectiveness of the proposed SwarmFusionDNN using the Br35H dataset, which comprises two distinct classes: tumor and normal. A comparison is made between the classification results of individual models, including MobileNet, DenseNet121, MobileNetV2, and Xception, alongside the SwarmFusionDNN ensemble model. The confusion matrix provides insights into the classification performance of each model. For MobileNet, out of 600 samples, 4 samples were misclassified. Similarly, DenseNet121 had 17 misclassified samples, MobileNetV2 had 32 misclassified samples, and Xception had 18 misclassified samples. However, our SwarmFusionDNN ensemble model demonstrated superior performance, with only 2 misclassified samples out of the 600. Fig. 8 depicts the confusion matrix, highlighting the reduction in misclassifications achieved by our ensemble architecture. Clearly, the ensemble approach successfully harnessed the individual strengths of models, resulting in a notable enhancement in overall classification accuracy. These results indicate that our proposed ensemble architecture outperforms individual models and significantly reduces the misclassification rate on the Br35H dataset.

Table 4 presents the performance evaluation results on the Br35H dataset. The performance evaluation results on the Br35H dataset indicate that all models achieved high accuracy rates, with SwarmFusionDNN exhibiting the highest accuracy of 99.66 %. This indicates the effectiveness of the ensemble model in improving classification performance. SwarmFusionDNN also achieved high sensitivity, precision, and F1-score, indicating its ability to correctly identify positive instances, minimize false positives, and achieve a balanced trade-off between precision and recall. The superior performance of SwarmFusionDNN can be attributed to the utilization of the PSO algorithm and the classifier error rate as the objective function during the ensemble optimization process. PSO enables the algorithm to efficiently search for optimal weights for each base model, allowing SwarmFusionDNN to leverage the strengths of individual models and effectively combine their predictions. Through the minimization of the classifier error rate, SwarmFusionDNN assigns greater weights to highly accurate models and lesser weights to less accurate ones. This strategy contributes to an enhanced overall classification accuracy. This demonstrates the advantage of incorporating swarm intelligence-based optimization techniques and using the classifier error rate as the objective function in ensemble modeling for improved performance.

Fig. 9 displays the ROC curves for the different models evaluated on the Br35H dataset. A higher area under the curve (AUC) indicates better classification performance. In the ROC curve analysis, SwarmFusionDNN achieved the highest AUC of 0.9998, indicating excellent

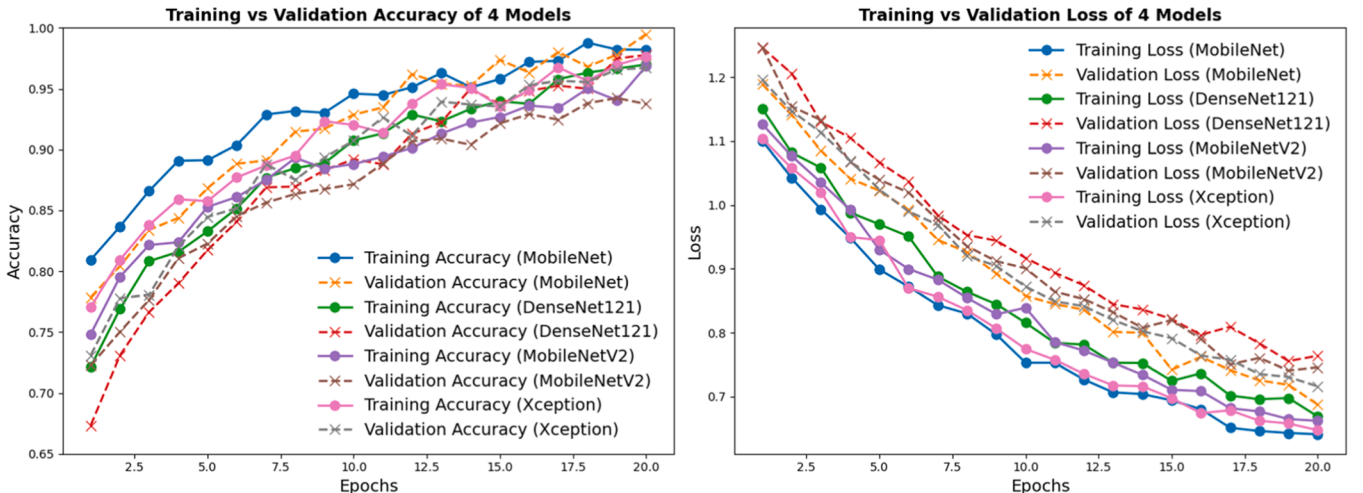


Fig. 7. Training and validation accuracy/loss curves on Br35H dataset.

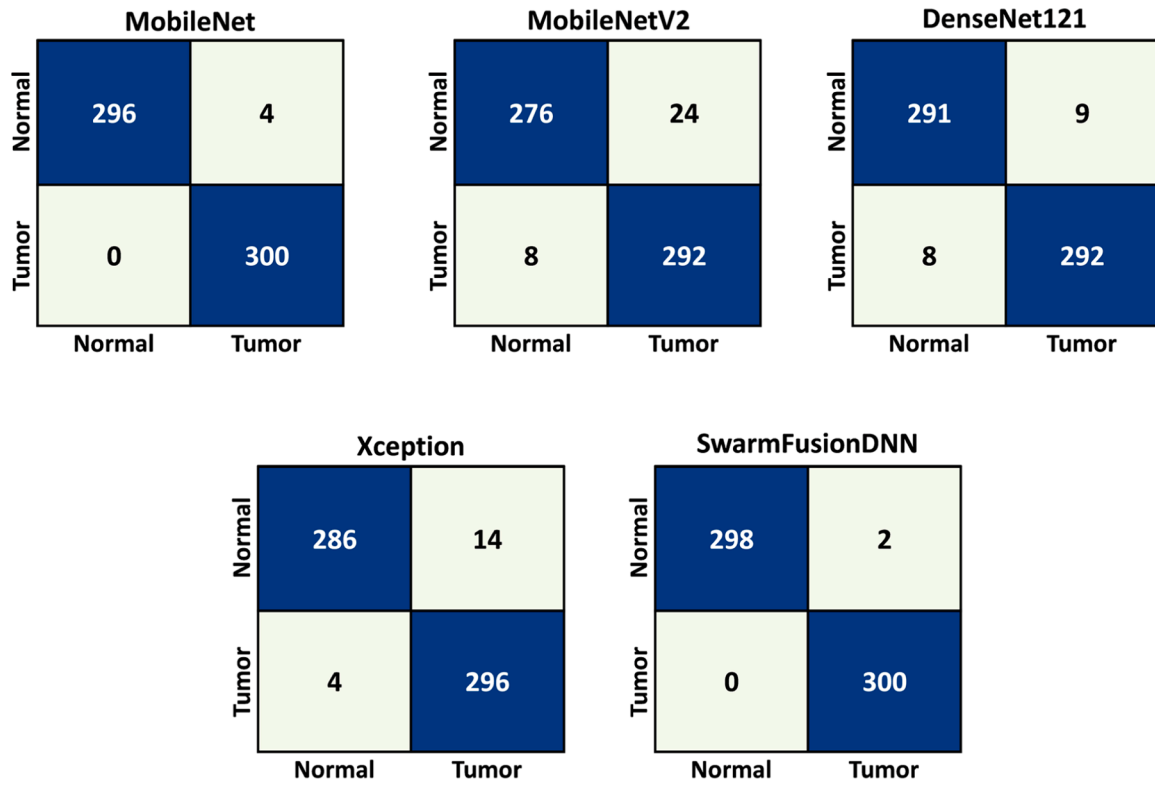


Fig. 8. Confusion matrix depicting the classification results of individual models and the SwarmFusionDNN ensemble model on the Br35H dataset.

Table 4
Performance evaluation results on the Br35H dataset.

Models	ACCU	SENS	PREC	F1
MobileNet	99.33 %	100 %	98.68 %	99.33 %
DenseNet121	97.16 %	97.33 %	97.00 %	97.17 %
MobileNetV2	94.66 %	97.33 %	92.40 %	94.80 %
Xception	97.00 %	98.66 %	95.48 %	97.04 %
SwarmFusionDNN	99.66 %	100 %	99.33 %	99.66 %

discrimination ability. The high AUC value suggests that the ensemble model is capable of accurately distinguishing between the classes in the dataset. The ROC curve and AUC analysis hold significant importance in the medical field, as they provide valuable insights into the diagnostic accuracy and reliability of a classification model.

Fig. 10 illustrates the comparison of error rates among different models. It is evident from the results that SwarmFusionDNN achieved the lowest error rate of 0.34 %, indicating its superior performance in classification accuracy. This highlights the effectiveness of the proposed SwarmFusionDNN ensemble architecture, which leverages the PSO algorithm and classifier error rate as the objective function to optimize the weights of the individual models. The significantly lower error rate of SwarmFusionDNN demonstrates its ability to make more accurate predictions compared to the other models.

4.4. Performance evaluation on Figshare dataset

This section presents the performance evaluation of the proposed ensemble architecture on the Figshare dataset, which contains three classes: glioma, meningioma, and pituitary. The models, including MobileNet, DenseNet121, MobileNetV2, Xception, and SwarmFusionDNN, were evaluated using confusion matrices to assess the misclassifications. The confusion matrices for each individual model are as follows: MobileNet (47/622), DenseNet121 (32/622), MobileNetV2 (49/622), Xception (69/622), and SwarmFusionDNN (24/622). Clearly,

the ensemble model, SwarmFusionDNN, successfully reduced misclassifications when compared to the individual models. This underscores the efficacy of the ensemble approach in enhancing the overall classification accuracy. Fig. 11 presents the confusion matrix for the Figshare dataset, providing a visual representation of the misclassifications. It can be observed that the SwarmFusionDNN model exhibits a lower number of misclassifications compared to the other models, further confirming its superior performance in accurately classifying the brain tumor images. Overall, the experimental results demonstrate that the ensemble approach, SwarmFusionDNN, effectively reduces misclassifications and improves the classification accuracy on the Figshare dataset.

Table 5 provides the performance metrics for each individual model on the Figshare dataset. The results show that MobileNet achieved an accuracy of 92.62 %, sensitivity of 92.16 %, precision of 91.63 %, F1-score of 91.88 %, and specificity of 96.23 %. DenseNet121 outperformed MobileNet with an accuracy of 94.85 %, sensitivity of 94.43 %, precision of 94.50 %, F1-score of 94.47 %, and specificity of 95.85 %. MobileNetV2 achieved an accuracy of 92.44 %, sensitivity of 91.65 %, precision of 91.65 %, F1-score of 91.64 %, and specificity of 96.03 %. Xception exhibited lower performance, with an accuracy of 88.90 %, sensitivity of 88.07 %, precision of 87.72 %, F1-score of 87.88 %, and specificity of 94.37 %. Notably, SwarmFusionDNN achieved the highest performance among all the models, with an accuracy of 96.14 %, sensitivity of 95.85 %, precision of 95.68 %, F1-score of 95.76 %, and specificity of 98.07 %. This remarkable performance can be attributed to the utilization of the PSO algorithm and the classifier error rate as the objective function in optimizing the weights of the ensemble.

Table 6 presents the class-wise AUC and F1-scores for three tumor types across five models. The proposed SwarmFusionDNN outperforms all baseline models in all classes, achieving the highest AUC and F1-score, particularly excelling in Pituitary tumor detection (AUC: 99.74 %, F1: 97.93 %). Among baselines, DenseNet121 and MobileNetV2 show competitive performance, while Xception lags slightly, especially on Meningioma. These results highlight the superior accuracy and

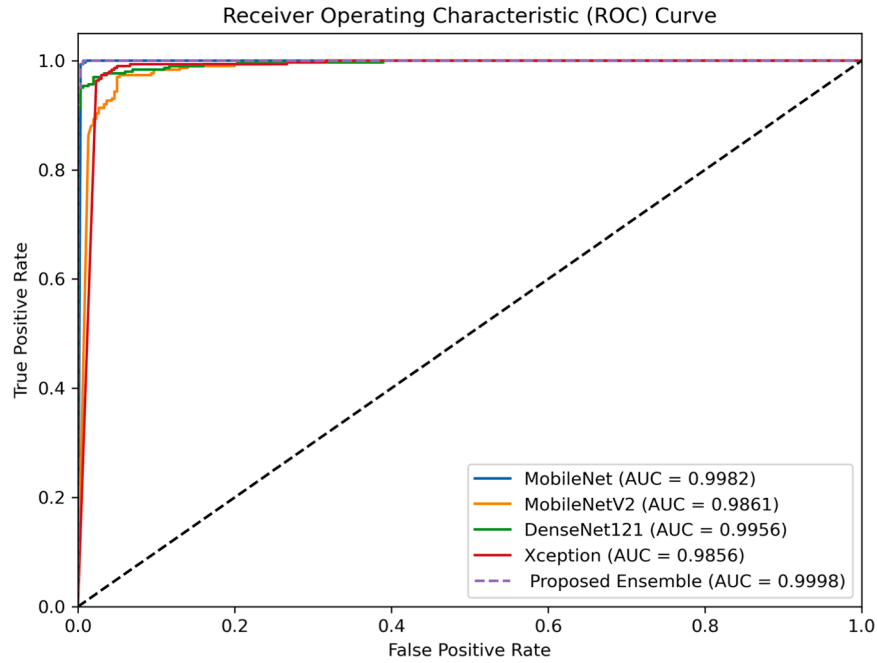


Fig. 9. ROC curve for different models evaluated on the Br35H dataset.

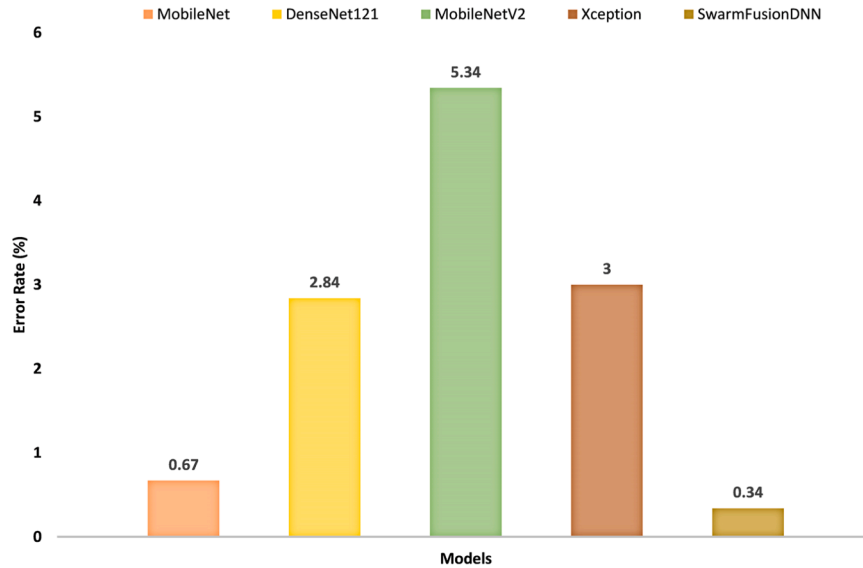


Fig. 10. Error rate comparison among different models.

generalizability of SwarmFusionDNN.

Fig. 12 illustrate the ROC curve for three-class classification on the Figshare dataset. The plot illustrates the performance of different models, including SwarmFusionDNN, in terms of AUC. SwarmFusionDNN achieved the highest AUC of 0.9946, indicating its superior ability to discriminate between glioma, meningioma, and pituitary tumors. The ROC curve analysis is particularly important in the medical field for assessing the diagnostic accuracy of classification models.

Fig. 13 show the comparison of error rates for different models on the Figshare dataset. The error rate represents the percentage of misclassified samples. The results demonstrate that SwarmFusionDNN achieves a significantly lower error rate compared to the individual models. This highlights the effectiveness of the swarm intelligence-based PSO algorithm and the use of classifier error rate as the objective function in optimizing the ensemble weights. The exceptional performance of SwarmFusionDNN underscores its capacity to harness the

combined intelligence of the ensemble, concluding in predictions that are both more accurate and dependable.

4.5. Performance evaluation on Bangladesh brain cancer dataset

This section evaluates the performance of the proposed ensemble architecture on the Bangladesh Brain Cancer Dataset, which includes three classes: glioma, meningioma, and tumor. The models—MobileNet, DenseNet121, MobileNetV2, Xception, and SwarmFusionDNN—were assessed using confusion matrices to highlight misclassifications. The misclassifications for each model are as follows: MobileNet (29/1212), DenseNet121 (46/1212), MobileNetV2 (29/1212), Xception (54/1212), and SwarmFusionDNN (15/1212). The ensemble model, SwarmFusionDNN, outperformed the individual models by reducing misclassifications. Fig. 14 shows the confusion matrix, illustrating the SwarmFusionDNN model's superior performance in minimizing

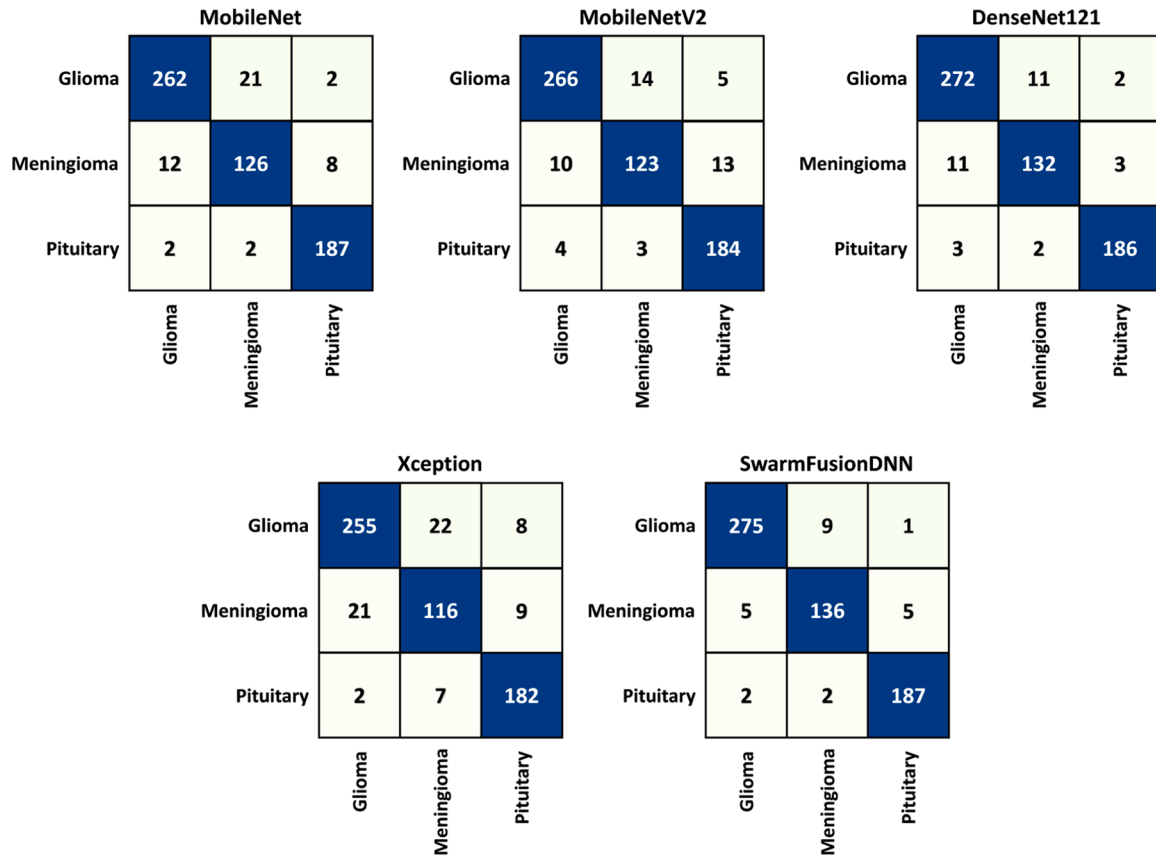


Fig. 11. Confusion matrix depicting the classification results of individual models and the SwarmFusionDNN ensemble model on the figshare dataset.

Table 5

Performance metrics of the models on the Figshare dataset.

Models	ACCU	SENS	PREC	F1	SPEC
MobileNet	92.44 %	91.47 %	92.04 %	91.74 %	96.23 %
DenseNet121	94.85 %	94.43 %	94.50 %	94.47 %	95.85 %
MobileNetV2	92.12 %	91.31 %	91.30 %	91.27 %	96.03 %
Xception	88.90 %	88.07 %	87.72 %	87.88 %	94.37 %
SwarmFusionDNN	96.14 %	95.85 %	95.68 %	95.76 %	98.07 %

Table 6

Class-wise AUC and F1-scores (%) for three tumor types.

Model	Class	AUC	F1-score
MobileNet	Glioma	98.49 %	93.40 %
	Meningioma	97.23 %	85.42 %
	Pituitary	99.58 %	96.39 %
DenseNet121	Glioma	98.65 %	95.27 %
	Meningioma	98.50 %	90.72 %
	Pituitary	99.50 %	97.38 %
MobileNetV2	Glioma	98.97 %	94.16 %
	Meningioma	98.08 %	86.01 %
	Pituitary	99.27 %	93.64 %
Xception	Glioma	97.14 %	90.59 %
	Meningioma	94.81 %	79.73 %
	Pituitary	98.01 %	93.33 %
SwarmFusionDNN	Glioma	99.47 %	97.00 %
	Meningioma	98.79 %	92.10 %
	Pituitary	99.74 %	97.93 %

misclassifications. These results demonstrate that the ensemble approach enhances classification accuracy and effectively reduces errors.

Table 7 shows the performance of five models on the Bangladesh

Brain Cancer Dataset. MobileNet and MobileNetV2 both achieved an accuracy of 97.61 %, with consistent values for sensitivity, precision, and F1-score (97.61 %). DenseNet121 performed slightly lower with an accuracy of 96.20 %, while Xception had the lowest accuracy at 95.54 %. The SwarmFusionDNN ensemble model outperformed all individual models, achieving an accuracy of 98.76 % and higher values across sensitivity, precision, and F1-score. This demonstrates the effectiveness of the ensemble approach in improving classification accuracy.

Fig. 15 presents the ROC curves for various models, highlighting their classification performance based on AUC values. Among them, SwarmFusionDNN achieved the highest AUC of 0.9991, demonstrating exceptional capability in distinguishing between glioma, meningioma, and tumor classes. This ROC analysis underscores the model's strong diagnostic accuracy, which is particularly valuable in medical applications. Fig. 16 compares the error rates of various models, showing that SwarmFusionDNN consistently outperforms the individual models with a notably lower error rate. This outcome underscores the effectiveness of employing the PSO-based swarm intelligence algorithm and using classifier error rate as the optimization objective for determining ensemble weights.

4.6. Statistical analysis

To statistically validate the performance improvements achieved by our proposed SwarmFusionDNN model over the baseline models, we conducted the McNemar test. This non-parametric test is specifically used to evaluate the significance of differences in classification performance between two models on paired nominal data. The McNemar test was performed for each baseline model—MobileNet, DenseNet121, MobileNetV2, and Xception—across three datasets: Br35H, Figshare, and the Bangladesh Brain Cancer dataset. As shown in Table 8, all p-values are significantly lower than the standard significance level of

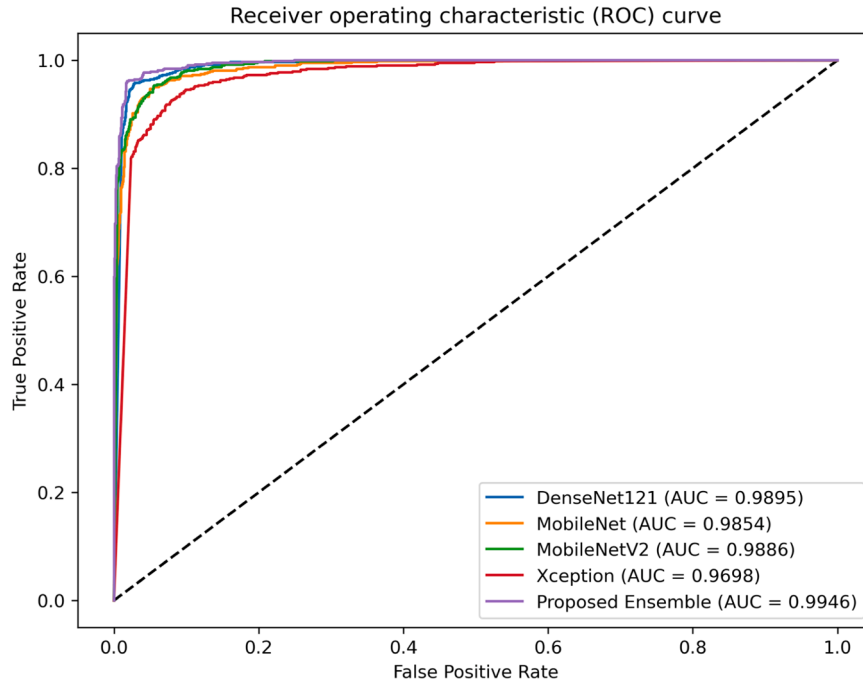


Fig. 12. ROC curve for three-class classification on the Figshare dataset.

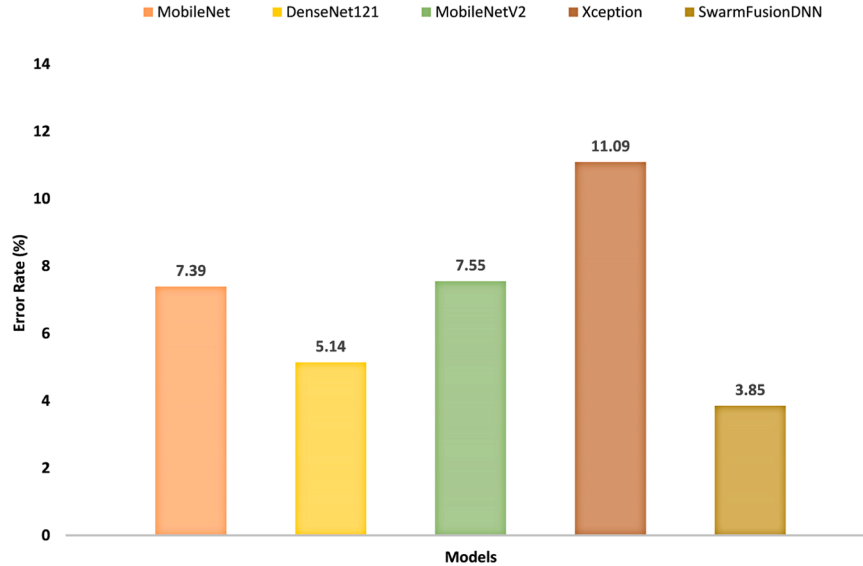


Fig. 13. Comparison of error rates for different models on the Figshare dataset.

0.05, indicating that the performance differences between SwarmFusionDNN and each individual model are statistically significant. For the Br35H dataset, the smallest p-value ($2.68e-38$) was observed when comparing SwarmFusionDNN with MobileNet, while all other comparisons also showed strong statistical significance ($p < 1e-10$). Similarly, on the Figshare dataset, extremely low p-values (as small as $1.32e-84$) reaffirm the superior and consistent performance of the proposed model. Lastly, for the Bangladesh Brain Cancer dataset, the p-values ranged from $3.28e-12$ to $1.72e-18$, again confirming that the performance improvements are not due to random chance. These results collectively confirm that SwarmFusionDNN offers statistically significant improvements in classification accuracy over all four baseline models across all evaluated datasets.

4.7. Analysis of misclassified samples

To better understand the limitations of our model and address potential sources of error, we present a focused analysis of misclassified samples across three benchmark datasets: Br35H, FigShare, and Bangladesh Brain Cancer MRI. Fig. 17 illustrates representative misclassified cases with both true and predicted labels. Notably, two normal samples from the Br35H dataset were falsely identified as tumor, potentially due to subtle artifacts or abnormal intensity regions mimicking pathological features. In the FigShare dataset, glioma and pituitary tumors were incorrectly predicted as meningioma, suggesting confusion between tumor classes with overlapping anatomical locations or similar texture patterns. Additionally, a meningioma case was misclassified as pituitary, indicating the model's occasional difficulty in differentiating tumors with overlapping radiological features. For the

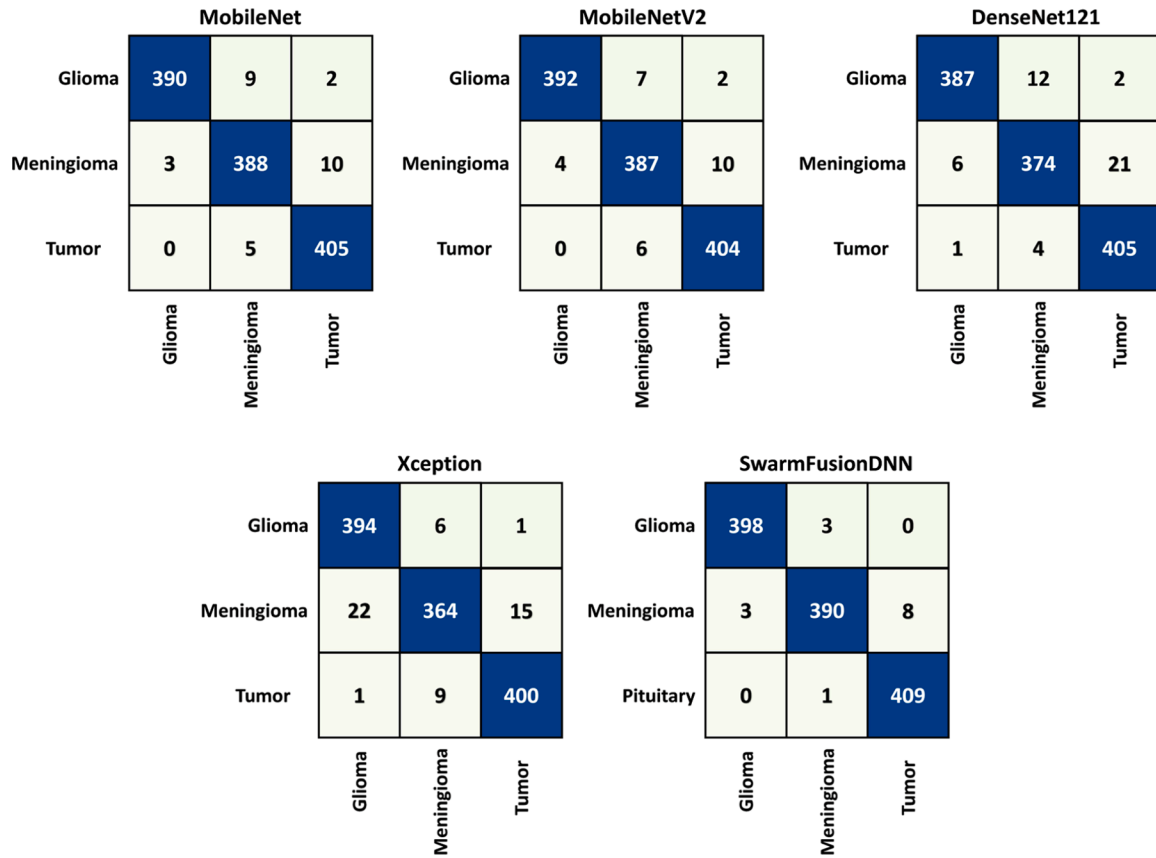


Fig. 14. Confusion matrices of the individual models and the SwarmFusionDNN ensemble on the Bangladesh Brain Cancer Dataset.

Table 7

Performance metrics of the models.

Models	ACC	SENS	PREC	F1	SPEC
MobileNet	97.61 %	97.62 %	97.61 %	97.61 %	98.80 %
DenseNet121	96.20 %	96.24 %	96.20 %	96.20 %	98.09 %
MobileNetV2	97.61 %	97.61 %	97.61 %	97.61 %	98.80 %
Xception	95.54 %	95.56 %	95.54 %	95.52 %	97.73 %
SwarmFusionDNN	98.76 %	98.77 %	98.75 %	98.75 %	99.37 %

Bangladesh dataset, gliomas and other tumors were repeatedly predicted as meningiomas, showing a tendency to overgeneralize toward the meningioma class—likely due to class imbalance or insufficient diversity in the glioma and generic “tumor” samples. Overall, most errors occurred in cases with low contrast, ambiguous tumor boundaries, or atypical anatomical presentations, highlighting the need for enhanced data diversity and augmentation strategies. This analysis underscores areas for improvement and motivates future work focusing on hard-case mining and advanced attention mechanisms.

4.8. Analysis of model sensitivity to noise and artifacts

To evaluate the robustness and generalizability of the proposed model, we introduced salt and pepper noise to the MRI images and analyzed the model’s performance under noisy conditions. As shown in Fig. 18, representative samples from each tumor category (glioma, meningioma, and pituitary) are presented, with predictions from DenseNet121, MobileNet, MobileNetV2, Xception, and the proposed model. It is evident that the proposed model demonstrates higher resilience to noise, consistently predicting the correct class even when traditional models are misled. For example, in the first row, the MobileNet and Xception models misclassify the glioma sample, while the proposed

model accurately identifies it. Similarly, in the third row, several baseline models fail to classify the pituitary tumor correctly under noise, whereas the proposed model maintains high accuracy. This illustrates the model’s superior ability to extract discriminative features and remain robust in the presence of artifacts.

4.9. Computational efficiency of base models

To assess the real-world applicability of the proposed ensemble system, we evaluated the computational efficiency of each base model after integrating the ConvMixer block. This additional module enhances spatial feature extraction, contributing to improved performance, but also slightly impacts computational requirements. Table 9 provides a detailed summary of the number of parameters, model size (MB), floating-point operations per image (FLOPs), and inference time on the test set. MobileNet, known for its lightweight design, remained highly efficient even after ConvMixer integration. It comprised 4.44 million parameters, required 1.25 GFLOPs, and consumed only 27.71 MB of storage, completing inference in just 3.78 s. MobileNetV2 showed the lowest computational demand, with only 0.77 GFLOPs and a model size of 31.61 MB, taking 5.14 s for inference—demonstrating its robustness and speed in resource-constrained scenarios. DenseNet121, with its dense connectivity and moderate complexity, increased slightly in computational demand post-ConvMixer enhancement, requiring 5.80 GFLOPs and 43.87 MB of memory. However, it remains a solid choice for clinical tasks where both accuracy and speed are needed, completing testing in 16.14 s. Xception, the most complex base model, benefited significantly from the ConvMixer module but exhibited the highest computational cost. It contained 25.38 million parameters, required 9.55 GFLOPs, and occupied 138.05 MB, completing inference in 11.48 s. Despite its complexity, Xception adds high-capacity feature extraction, enriching the diversity of the ensemble.

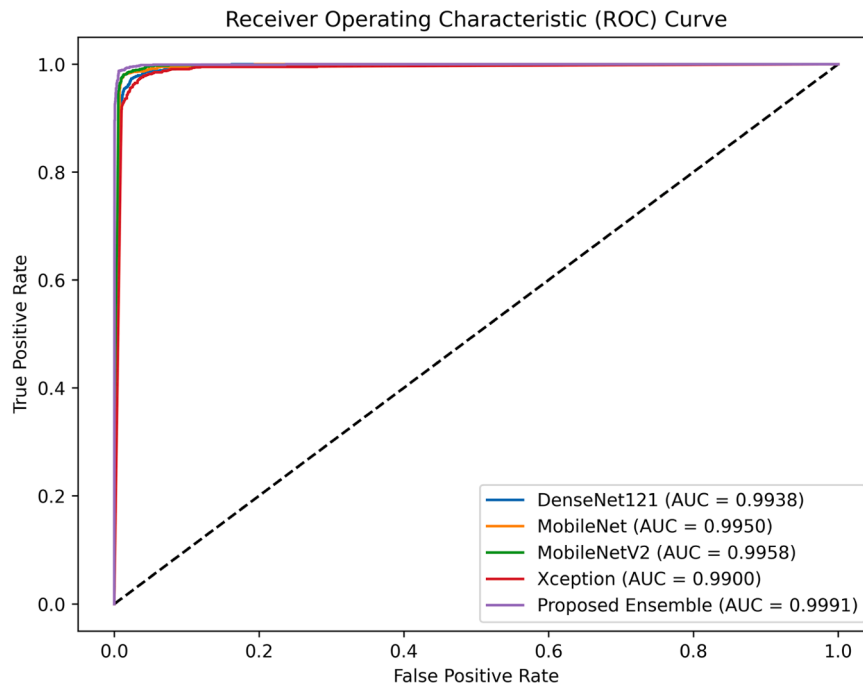


Fig. 15. ROC curve for different models.

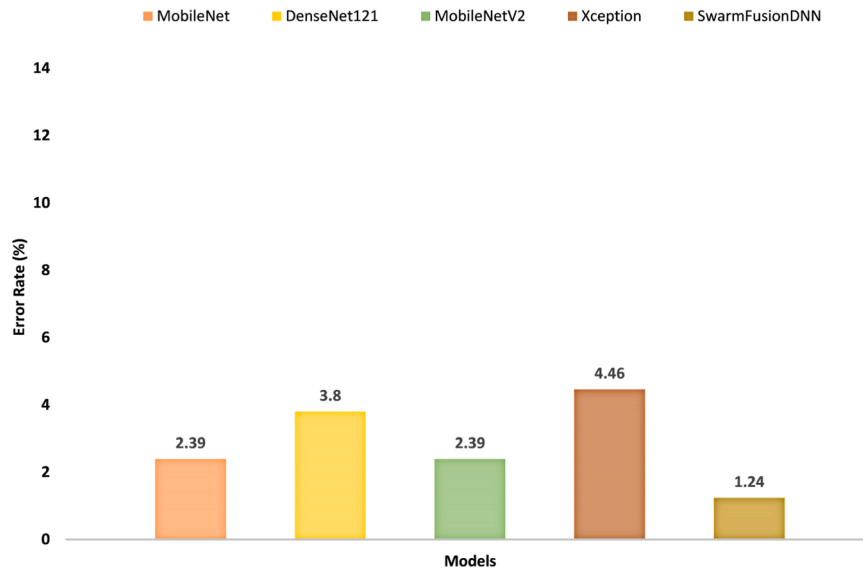


Fig. 16. Comparison of error rates across different models.

Table 8
p-values from the McNemar test.

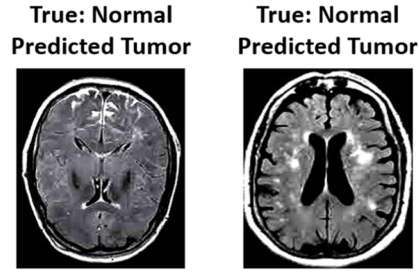
Dataset	Comparison Model	p-value
Br35H	MobileNet	2.68e-38
	DenseNet121	1.16e-11
	MobileNetV2	7.13e-11
	Xception	4.86e-11
	MobileNet	5.00e-36
Figshare	DenseNet121	1.21e-63
	MobileNetV2	5.24e-69
	Xception	1.32e-84
	MobileNet	3.28e-12
	DenseNet121	2.91e-12
Bangladesh Brain Cancer Dataset	MobileNetV2	2.97e-16
	Xception	1.72e-18

To further optimize these models for clinical deployment, techniques such as model quantization and pruning can be applied to reduce memory usage and inference time without significantly sacrificing accuracy. However, real-world deployment still poses challenges, including hardware limitations, integration with hospital systems, and maintaining performance across diverse patient data.

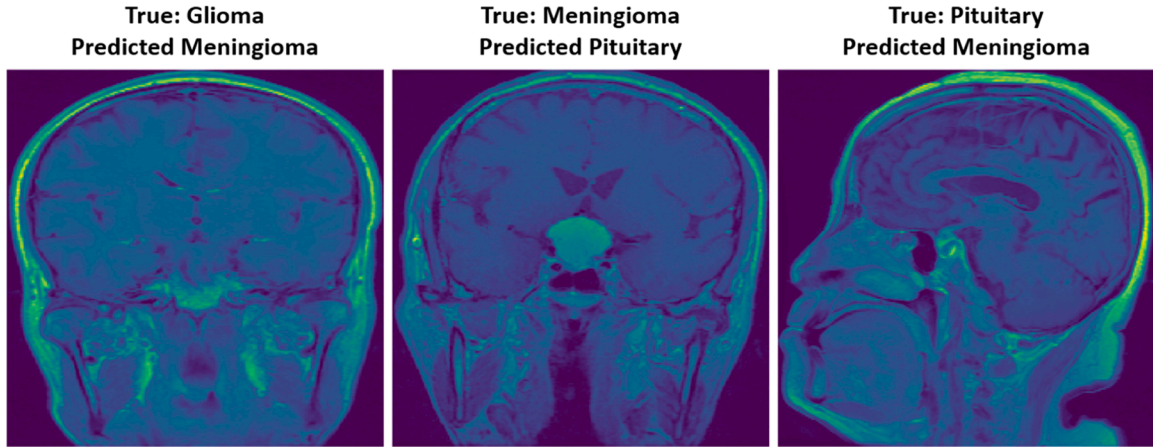
4.10. Exploring ensemble diversity: the influence of varying base model count

The impact of varying the number of base models in an ensemble configuration is evident from the performance metrics presented. The Table 10 displays the results of different combinations of base models, and it is clear that the addition of more base models generally leads to improved performance across multiple evaluation metrics. Starting with

Br35H Dataset



FigShare Dataset



Bangladesh Brain Cancer MRI Dataset

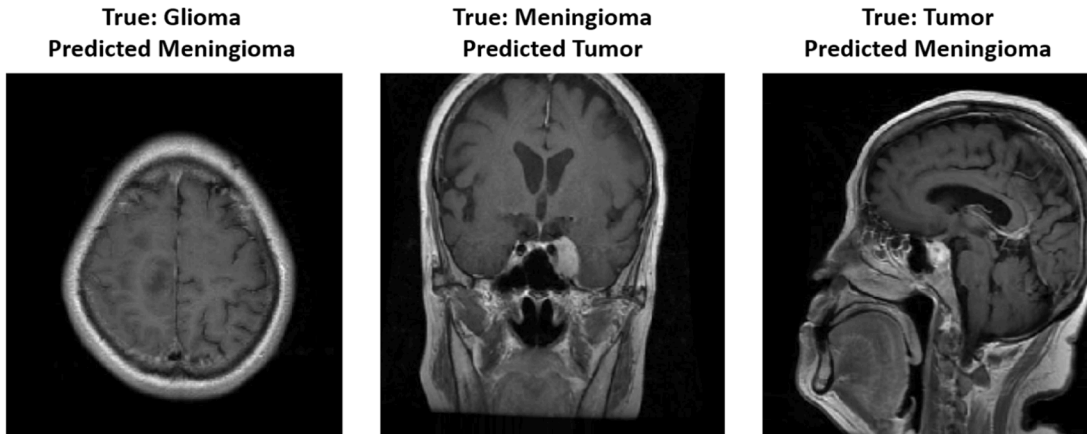


Fig. 17. Representative misclassified samples from the Br35H, FigShare, and Bangladesh Brain Cancer MRI datasets.

a combination of DenseNet121 and MobileNet, a respectable accuracy of 95.50 % is observed. However, when Xception is added to the ensemble, the accuracy increases to 95.99 %, indicating the benefit of diversity in model architectures. Furthermore, including MobileNetV2 alongside DenseNet121 and Xception maintains high accuracy at 95.82 %, emphasizing the robustness of the ensemble. Interestingly, the addition of MobileNetV2 to the combination of DenseNet121 and MobileNet leads to a slight drop in accuracy but maintains competitive precision and F1-scores. The most striking improvement is seen when all four base models are combined in the SwarmFusionDNN, achieving an accuracy of 96.14 %. This indicates that a larger ensemble of diverse models can effectively boost performance, achieving higher accuracy, sensitivity,

precision, and F1-scores. In conclusion, the results demonstrate that the number and diversity of base models in an ensemble can significantly impact the overall performance, highlighting the importance of carefully designing ensemble configurations to achieve optimal results in machine learning tasks.

4.11. Comparative analysis with pre-trained models

The SwarmFusionDNN, as proposed in our study, exhibited strong performance when assessed using various metrics tailored to our task. Nevertheless, to ascertain its effectiveness, it's essential to compare its performance against other pre-trained models designed for brain tumor

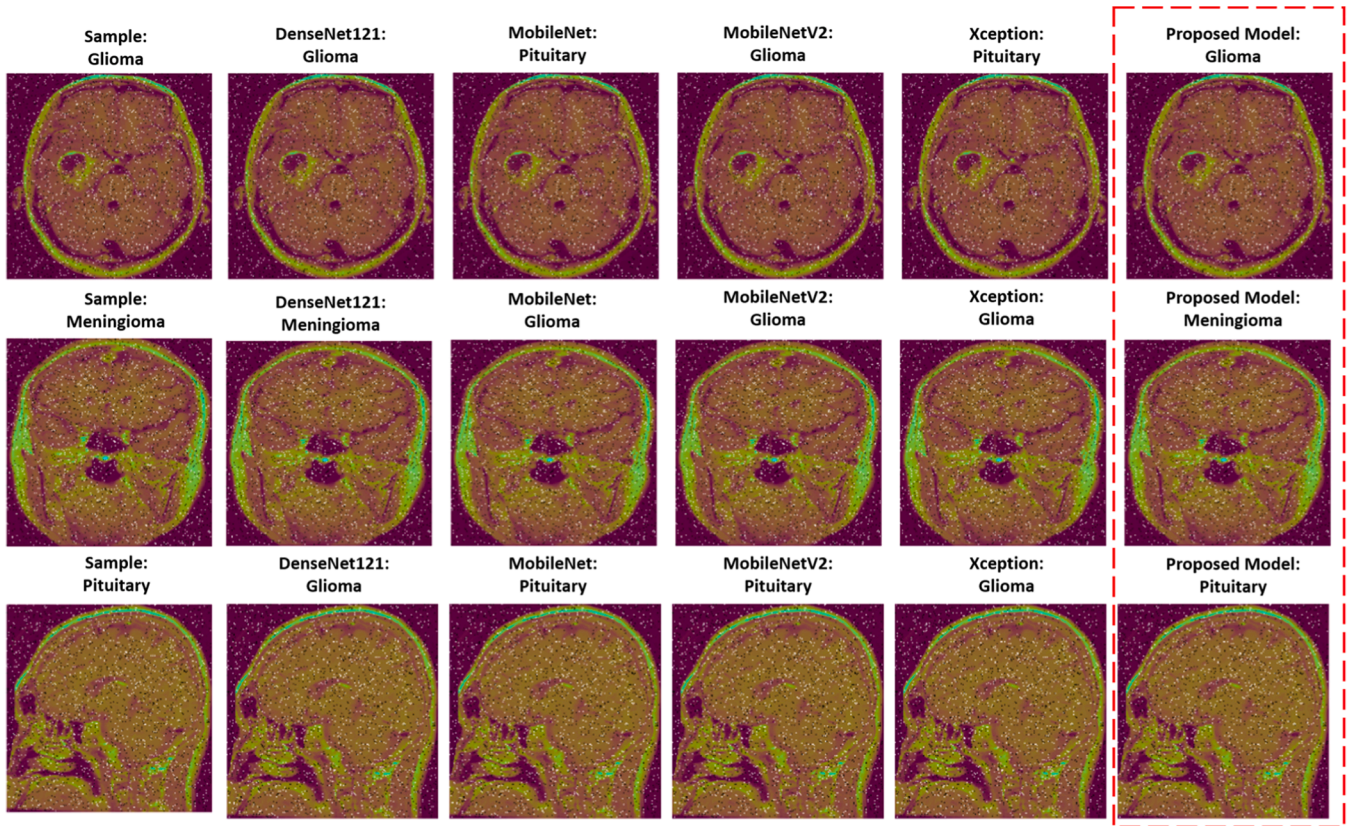


Fig. 18. Comparison of classification performance under salt and pepper noise for different models.

Table 9

Computational and Inference Performance of Base Models with ConvMixer Block.

Model	Parameters (M)	Model Size (MB)	FLOPs (G)	Test Time (s)
MobileNet	4.44	27.71	1.25	3.78
DenseNet121	8.25	43.87	5.80	16.14
MobileNetV2	4.10	31.61	0.77	5.14
Xception	25.38	138.05	9.55	11.48

Table 10

Performance metrics for different ensemble configurations with varying base model counts.

Ensemble Configuration	ACC	SENS	PREC	F1
DenseNet121 + MobileNet	95.50	95.49	95.52	95.51
	%	%	%	%
DenseNet121 + MobileNet + Xception	95.99	95.98	95.97	95.98
	%	%	%	%
DenseNet121 + MobileNetV2 + Xception	95.82	95.81	95.82	95.81
	%	%	%	%
DenseNet121 + MobileNet + MobileNetV2	95.81	95.85	95.81	95.82
	%	%	%	%
DenseNet121 + MobileNet + MobileNetV2 + Xception (SwarmFusionDNN)	96.14	96.14	96.15	96.14
	%	%	%	%

diagnosis. Table 11 presents comprehensive comparison between SwarmFusionDNN and other pre-trained models. Our findings demonstrate that the SwarmFusionDNN outperforms individual models, providing strong empirical evidence supporting its superior reliability and robustness. The results of SwarmFusionDNN are attributed to its ability to effectively fuse information from multiple CNN models through a weighted average ensemble technique, optimizing model

Table 11

Performance comparison of SwarmFusionDNN with other pre-trained models for brain tumor diagnosis.

Models	ACC	SENS	PREC	F1	SPEC
MobileNet	92.44	91.47	92.04	91.74	96.23
	%	%	%	%	%
DenseNet121	94.85	94.43	94.50	94.47	95.85
	%	%	%	%	%
MobileNetV2	92.12	91.31	91.30	91.27	96.03
	%	%	%	%	%
Xception	88.90	88.07	87.72	87.88	94.37
	%	%	%	%	%
DenseNet201	85.20	84.14	83.14	83.51	92.88
	%	%	%	%	%
InceptionV3	78.77	74.50	77.68	74.59	88.65
	%	%	%	%	%
InceptionResNetV2	76.52	71.37	73.19	70.24	88.05
	%	%	%	%	%
NasNetMobile	81.51	77.43	80.61	77.01	90.05
	%	%	%	%	%
ResNet50V2	79.09	76.63	77.44	76.76	89.60
	%	%	%	%	%
Proposed SwarmFusionDNN	96.14	96.14	96.15	96.14	98.07
	%	%	%	%	%

performance and leveraging diverse insights. Additionally, the incorporation of PSO in SwarmFusionDNN plays a crucial role in fine-tuning model weights, enabling it to maximize the utilization of individual CNN models' strengths, ultimately contributing to its excellent performance. The results of our research serve as concrete evidence that ensemble techniques can effectively exploit the complementary information offered by multiple models, leading to enhanced performance.

4.12. Grad-CAM visualization

Grad-CAM visualization [51] is of significant importance in the field of medical diagnosis. It provides valuable insights into the regions of interest in medical images that contribute most to the predictions made by DL models. By highlighting these salient regions, Grad-CAM helps clinicians and researchers understand the reasoning behind the model's decision-making process. Fig. 19 presents Grad-CAM visualizations for three brain tumor cases from the Br35H dataset. Each row corresponds to a different patient image with a tumor clearly marked in the original image. For each input image, the Grad-CAM outputs of four different models are shown alongside their heatmaps. The Grad-CAM heatmaps clearly highlight regions that strongly align with the ground truth tumor locations marked in the input image. All four models consistently activate around the tumor-affected areas, indicating that they are correctly attending to the pathological regions contributing to the classification. This strong alignment with the annotated tumor regions enhances the model's transparency and diagnostic reliability, demonstrating that the ensemble learns and localizes relevant features effectively. Fig. 20 shows Grad-CAM visualizations for three different brain tumor types from the Figshare dataset. Each row corresponds to one tumor type, with the input image followed by Grad-CAM outputs from the models. Grad-CAM outputs highlight discriminative regions within each tumor type, and these regions closely overlap with the visible tumor masses in the original images. Despite the diversity in tumor shapes and locations across classes, the models effectively localize the tumor zones, suggesting generalizability and robustness in feature detection. The consistent focus across models and tumor types strengthens confidence in the ensemble model's ability to accurately classify and localize different brain tumors. The Grad-CAM maps clearly depict that the models consistently focus on the tumor regions, underscoring the ensemble model's ability to accurately identify and localize brain tumors. This visual evidence further strengthens the reliability and robustness of the ensemble model's predictions, making it a valuable tool for precise and effective brain tumor diagnosis.

4.13. Ablation study

To assess the impact of integrating the ConvMixer block into the base models used to construct our SwarmFusionDNN, it is crucial to conduct a comparative analysis of model performance with and without the

ConvMixer block. This section presents a performance evaluation of these models to determine the effectiveness of the ConvMixer block's inclusion. The experiments were conducted on the Figshare Brain MRI dataset, which comprises three classes of brain tumors. Table 12 showcases the performance results of these models. To justify the integration of the ConvMixer block into the base models used in SwarmFusionDNN, it is essential to highlight its architectural advantages over commonly used enhancements such as Squeeze-and-Excitation (SE) blocks and the Convolutional Block Attention Module (CBAM). SE and CBAM primarily focus on refining channel-wise and spatial feature representations through attention mechanisms. In contrast, ConvMixer combines depthwise separable convolutions with large kernel convolutions in a patch-based manner, enabling effective spatial mixing and feature extraction with lower computational complexity. This design makes ConvMixer particularly suitable for medical image analysis tasks like brain tumor classification, where capturing fine-grained local and global structures is crucial. Its simplicity and strong performance across different backbone networks make it a compelling choice for enhancing SwarmFusionDNN.

The Benefits of the ConvMixer Block:

Enhanced Accuracy: As demonstrated in Table 12, incorporating the ConvMixer block consistently leads to improved model accuracy across various base models. For instance, the accuracy of the DenseNet121 model significantly increased from 87.90 % to 94.85 % when the ConvMixer block was added. This indicates that the ConvMixer block helps the models capture more complex features and make better predictions.

Generalization: ConvMixer's effectiveness across different base models, such as MobileNet, DenseNet121, MobileNetV2, and DenseNet201, highlights its ability to enhance the performance of a wide range of architectures. This suggests that the ConvMixer block contributes to better model generalization and can be considered a valuable addition to various neural network architectures.

Improved Learning: The ConvMixer block is particularly beneficial when dealing with complex datasets and tasks. It allows models to learn richer representations, making them more capable of handling intricate patterns and nuances in the data. This is evident in the substantial accuracy gains observed in models equipped with ConvMixer.

Ensemble Model Performance: When integrated into an ensemble model, the ConvMixer-enhanced base models are likely to contribute positively to the overall performance. This could result in a more robust

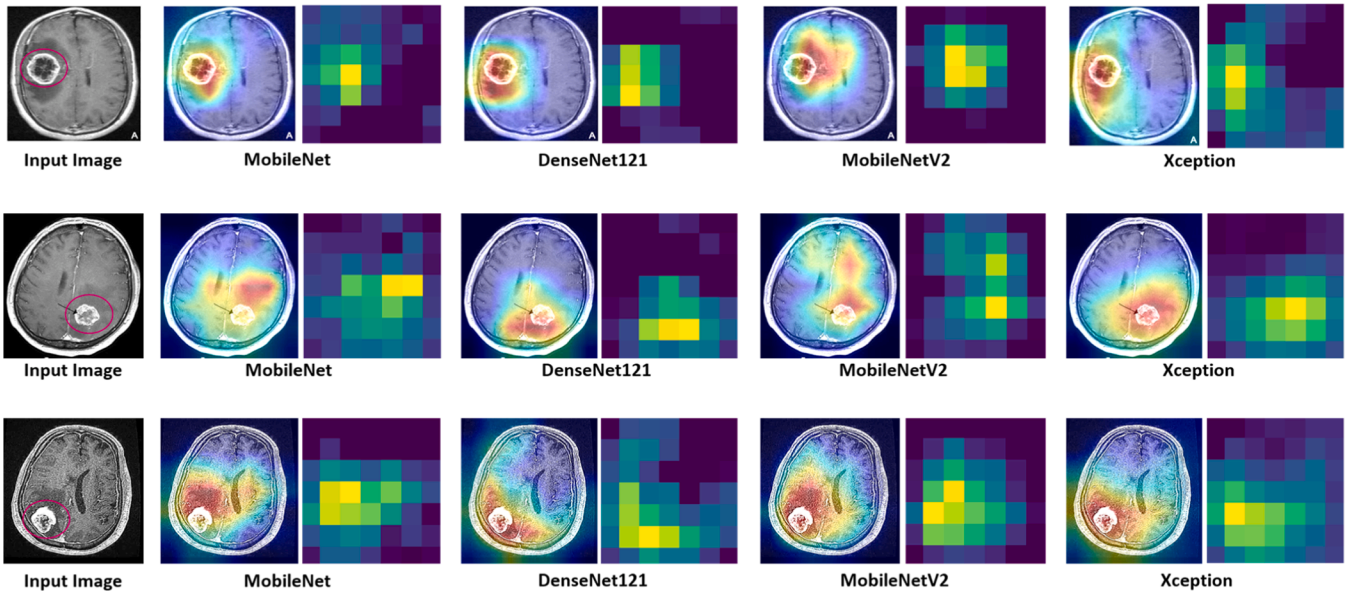


Fig. 19. Grad-CAM visualization on the BR35H dataset to detect the presence of brain tumors.

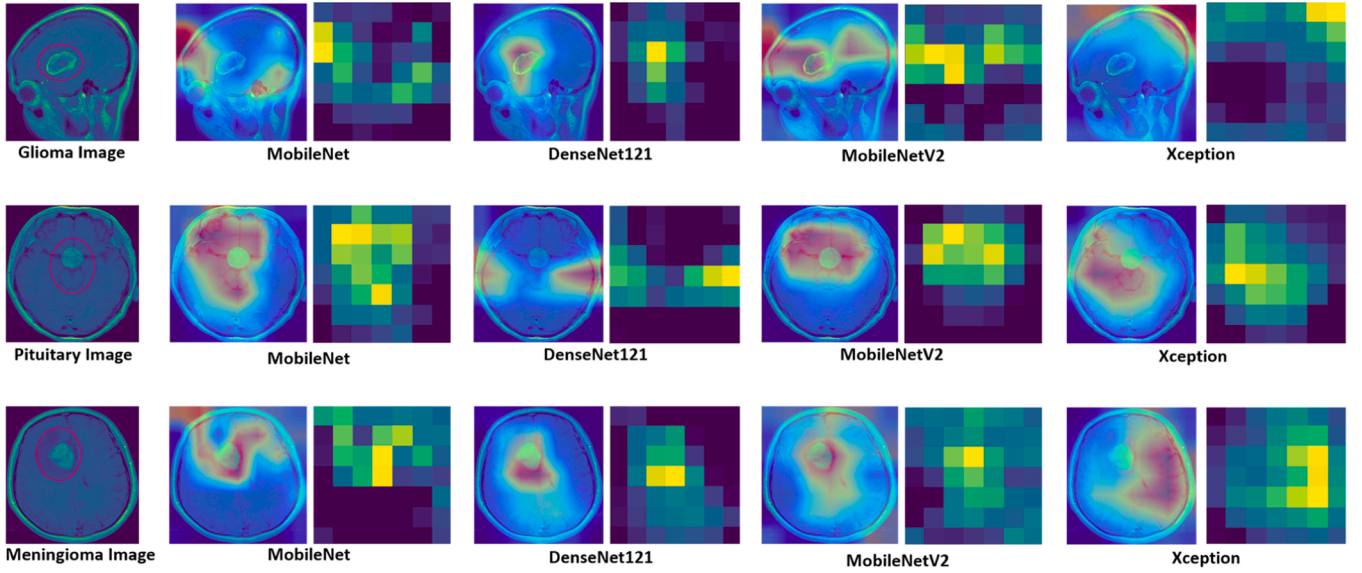


Fig. 20. Grad-CAM visualization on the Figshare dataset to identify three different brain tumor types.

Table 12

Performance comparison of base models with and without ConvMixer block on Figshare brain MRI dataset.

Models	Accuracy
MobileNet (Baseline)	91.32 %
MobileNet + ConvMixer	92.44 %
DenseNet121 (Baseline)	87.90 %
DenseNet121 + ConvMixer	94.85 %
MobileNetV2 (Baseline)	84.10 %
MobileNetV2 + ConvMixer	92.12 %
Xception (Baseline)	81.70 %
Xception + ConvMixer	88.90 %

and accurate ensemble model, making it a promising choice for ensemble-based learning strategies.

Table 13 presents the performance comparison of four base models (MobileNet, DenseNet121, MobileNetV2, and Xception) with and without augmentation, along with the ConvMixer block. The accuracy values show the impact of augmentation techniques on the models' performance. As observed, all models benefit from the addition of augmentation. MobileNet + ConvMixer, for example, shows an increase in accuracy from 91.96 % to 92.60 % after applying augmentation. Similarly, DenseNet121 + ConvMixer experiences a significant improvement, with accuracy rising from 92.28 % to 94.85 %. MobileNetV2 + ConvMixer sees a boost from 89.22 % to 92.44 %, and Xception + ConvMixer improves from 86.49 % to 88.90 %. These results highlight the effectiveness of augmentation in enhancing the generalization capability of the models, contributing to improved performance, especially in complex medical image classification tasks like brain MRI.

4.14. Comparison with other ensemble methods

In this section, the performance of the SwarmFusionDNN ensemble

Table 13

Performance comparison of base models with and without augmentation.

Models	Before Augmentation	After Augmentation
MobileNet + ConvMixer	91.96 %	92.44 %
DenseNet121 + ConvMixer	92.28 %	94.85 %
MobileNetV2 + ConvMixer	89.22 %	92.12 %
Xception + ConvMixer	86.49 %	88.90 %

model is compared with several other ensemble techniques on both the Br35H and Figshare datasets. The evaluation metric used is accuracy, which measures the overall classification performance of the models. Table 14 presents the comparison of ensemble methods on Br35H and Figshare datasets. To ensure a fair comparison, all ensemble methods were built using the same set of base models used in the proposed SwarmFusionDNN

On the Br35H dataset, the SwarmFusionDNN ensemble using PSO achieves the highest accuracy of 99.66 %. It outperforms other ensemble methods such as the average ensemble [52] (98.66 %), Condorcet's jury theorem [53] (98.83 %), Gompertz function [54] (98.67 %), weighted average ensemble with random search [11] (98.66 %), weighted average ensemble with ant colony optimization [55] (99.50 %), weighted average ensemble with grey wolf optimizer [56] (99.33 %), and stack ensemble [57] (98.67 %). Similarly, on the Figshare dataset, the SwarmFusionDNN achieves the highest accuracy of 96.14 %. It surpasses the performance of other ensemble methods including the average ensemble (95.49 %), Condorcet's jury theorem (95.17 %), Gompertz function (95.50 %), weighted average ensemble with random search (95.98 %), weighted average ensemble with ant colony optimization (95.81 %), weighted average ensemble with grey wolf optimizer (95.98 %), and stack ensemble (95.65 %). The comparison highlights the success of our SwarmFusionDNN ensemble model. The performance of SwarmFusionDNN can be attributed to two key factors. Firstly, its

Table 14

Comparison of ensemble methods on Br35H and Figshare datasets.

Dataset	Ensemble techniques	Accuracy
Br35H	Average ensemble	98.66 %
	Condorcet's jury theorem	98.83 %
	Gompertz function	98.67 %
	Weighted average ensemble (Random Search)	98.66 %
	Weighted average ensemble (Ant colony optimization)	99.50 %
	Weighted average ensemble (Grey wolf optimizer)	99.33 %
	Stack ensemble	98.67 %
Figshare	SwarmFusionDNN (PSO)	99.66 %
	Average ensemble	95.49 %
	Condorcet's jury theorem	95.17 %
	Gompertz function	95.50 %
	Weighted average ensemble (Random Search)	95.98 %
	Weighted average ensemble (Ant colony optimization)	95.81 %
	Weighted average ensemble (Grey wolf optimizer)	95.98 %
	Stack ensemble	95.65 %
	SwarmFusionDNN (PSO)	96.14 %

utilization of the PSO algorithm that enables to find the most optimal combination of weights, ensuring that the most accurate models within the ensemble have a more significant influence. Secondly, the incorporation of ConvMixer blocks into the base models enhances their capabilities, leading to higher accuracy and more robust feature extraction, which collectively contribute to the improved performance of SwarmFusionDNN. Overall, our SwarmFusionDNN ensemble model stands out as the top-performing approach on both datasets, demonstrating its effectiveness in brain tumor classification tasks.

4.15. Additional experiments

To validate the effectiveness of the proposed SwarmFusionDNN, additional experiments were conducted using a curated dataset [58] comprising 7023 MRI images divided into four classes. This dataset was created by integrating samples from renowned MRI sources, including the SARTAJ dataset [59], the Br35H dataset [60], and data from Cheng et al.'s study available on Figshare [61]. Table 15 summarizes the results of additional experiments conducted to further evaluate the performance of SwarmFusionDNN. These experiments were designed to assess its capability on a larger and more diverse curated MRI dataset. SwarmFusionDNN outperformed other models, reinforcing its reliability and consistency across different MRI datasets.

4.16. Comparison with existing methods

This study compares the proposed brain tumor detection method, SwarmFusionDNN, with several existing methods from the literature. The Table 16 summarizes the methods and accuracies reported by each method. The comparison with existing methods clearly demonstrates the effectiveness of our proposed SwarmFusionDNN ensemble model. By incorporating the PSO algorithm, our model achieves superior accuracy compared to other ensemble methods. Our proposed SwarmFusionDNN ensemble approach for brain tumor detection offers several significant advancements in accuracy and performance compared to single models and other ensemble techniques. One key aspect of our method is the incorporation of ConvMixer blocks into the architecture of the base models. This integration enhances the models' feature extraction capabilities, allowing them to effectively capture intricate spatial and channel-wise information from brain MRI scans. By leveraging ConvMixer blocks, our method improves the accuracy and reliability of tumor classification. Overall, the integration of ConvMixer blocks and the application of the PSO are key factors contributing to the superior performance of SwarmFusionDNN. These advancements enable our model to better capture complex spatial and channel-wise information and optimize the ensemble's weight assignment, leading to enhanced accuracy in brain tumor detection compared to other methods in the field.

4.17. Clinical application of swarmfusiondnn in brain tumor classification

Our research endeavors to advance the field of brain tumor classification through the development of SwarmFusionDNN, a cutting-edge model that combines swarm intelligence optimization and ensemble

Table 15

Performance comparison of SwarmFusionDNN with other models on the curated MRI dataset from additional experiments.

Models	ACCU	SENS	PREC	F1	SPEC
MobileNet	97.71 %	97.80 %	97.72 %	97.71 %	97.75 %
DenseNet121	97.55 %	95.63 %	97.55 %	97.54 %	97.68 %
MobileNetV2	97.40 %	97.49 %	97.40 %	97.41 %	97.53 %
Xception	97.78 %	97.86 %	97.78 %	97.78 %	97.89 %
SwarmFusionDNN	98.09 %	98.14 %	98.09 %	98.08 %	98.32 %

Table 16

Brain Tumor Detection Method Comparison of SwarmFusionDNN with Existing Methods.

Author	Methods	Br35H	Figshare Dataset
[15]	Ensemble	-	94.34 %
[62]	Hybrid	-	95.82 %
[63]	CNN	-	94.74 %
[64]	CNN	-	96.13 %
[65]	Hybrid	99.10 %	-
[66]	IVX16	-	96.94 %
[67]	TumorResNet	99.33 %	-
[27]	12 layer model	98.83 %	-
[68]	TL model	-	95.75 %
[69]	Majority voting	-	94.80 %
[70]	MCCNN	-	96.00 %
[71]	CNN AlexNet	-	94.70 %
[72]	Ensemble	98.48 %	-
[73]	Residual Model	99.02 %	-
Our Method	SwarmFusionDNN	99.66 %	96.14 %

techniques. By leveraging swarm intelligence, our model optimizes the classification process, enabling accurate identification of various types of brain tumors. The implications of our research extend beyond the laboratory, with numerous potential applications in the realm of neuro-oncology. The precise and reliable classification of brain tumors facilitated by SwarmFusionDNN holds the potential to significantly impact clinical practice and patient outcomes. Fig. 21 provides a visual representation of the potential application scenarios enabled by our SwarmFusionDNN. This illustration showcases the transformative potential of our model in facilitating accurate diagnoses, treatment planning, and monitoring of brain tumor patients. Through technical advancements and innovative approaches, this study aims to empower medical professionals with a powerful tool that enhances their ability to understand and address brain tumors effectively.

4.18. Empowering radiologists with accurate and timely classification

Our research enhances brain tumor diagnosis for radiologists and clinicians using MRI scans. By employing advanced algorithms, our approach ensures accurate classification, reducing misdiagnosis and enabling early tumor detection, which improves patient outcomes and supports timely intervention.

4.19. Precise treatment guidance

Our model excels in identifying the specific tumor type, providing clinicians with valuable insights for designing targeted treatment strategies. By leveraging this precise classification, clinicians can optimize treatment approaches, resulting in more effective and precise outcomes for patients.

4.20. Empowering informed decision-making with the model as a clinical support system

Our advanced model serves as a valuable tool in clinical settings, providing crucial support to healthcare providers for making well-informed decisions in patient management. By delivering accurate and dependable results in tumor classification, it significantly enhances clinical decision-making capabilities, enabling a higher level of precision and personalized care for patients.

4.21. Focused tumor region visualization for doctors and clinicians

Grad-CAM visualization highlights key tumor regions, allowing doctors to validate model predictions and understand its decision-making. This aids in treatment planning, enables targeted interventions, and improves communication with patients, empowering

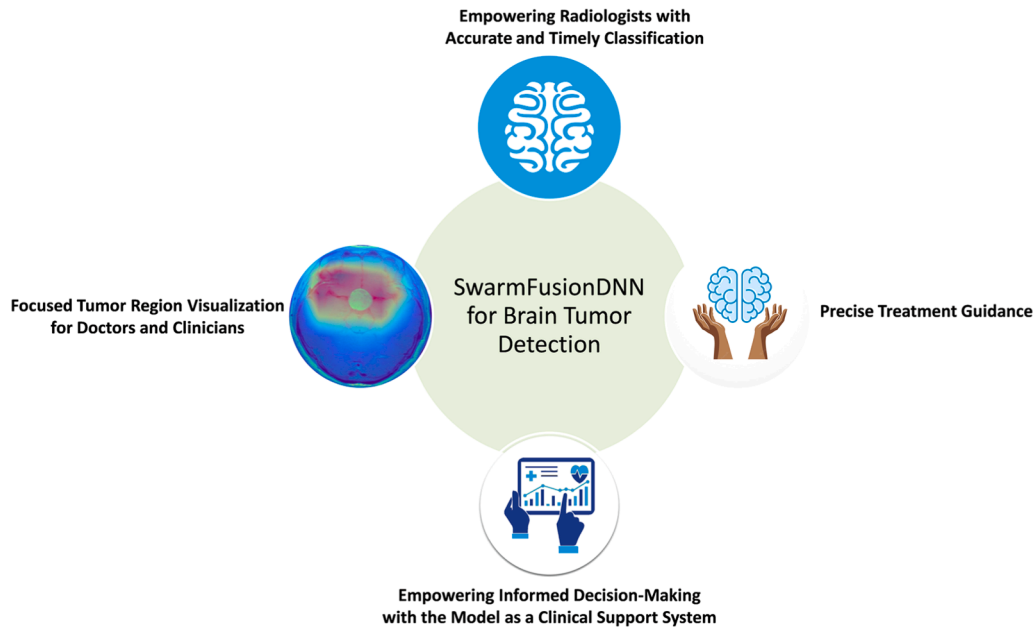


Fig. 21. Clinical Application of SwarmFusionDNN.

doctors to make informed decisions and deliver precise care.

4.22. Deployment challenges and considerations

While the clinical benefits of SwarmFusionDNN are significant, real-world deployment poses several challenges. These include:

- DICOM-compliant PACS integration, ensuring compatibility with hospital imaging infrastructure and standardized workflows.
- Real-time MRI processing requirements, where inference speed must be optimized to support time-sensitive diagnoses.
- Adaptation to clinical workflows, requiring user-friendly interfaces and minimal disruption to existing diagnostic procedures.
- IT infrastructure and data privacy compliance, including adherence to medical data standards such as HIPAA and GDPR.

4.23. Limitations and future direction

While the proposed SwarmFusionDNN model demonstrates superior performance in brain tumor detection, it is not without limitations. First, the model's reliance on multiple pre-trained networks and the ConvMixer blocks significantly increases its computational complexity. This may hinder its real-time applicability in clinical environments where computational resources are limited, particularly on edge devices or in resource-constrained settings. Second, the model's performance has been tested on a limited number of publicly available datasets (Br35H and Figshare). While these datasets provide valuable insights, the model's generalizability to other medical imaging modalities or more diverse brain tumor types has not been thoroughly explored. Additionally, the dataset imbalance in some cases may influence the model's ability to effectively detect less common tumor classes.

4.24. Future directions

To address the computational challenges, future work could focus on integrating model compression techniques, such as pruning and quantization, to reduce the model's complexity while maintaining performance. Moreover, exploring more lightweight architectures could improve the feasibility of deploying SwarmFusionDNN in real-time clinical settings. Further research should also investigate the extension

of SwarmFusionDNN to multi-modal medical imaging data, such as CT or PET scans, to enhance the model's diagnostic capabilities across various cancer types. Expanding the model to handle noisy, incomplete, or imbalanced data is another important area for future improvement, potentially through the integration of advanced data augmentation techniques or semi-supervised learning approaches.

Finally, evaluating the model on larger and more diverse datasets will be critical to ensuring its robustness and generalizability across different clinical scenarios.

5. Conclusion

In conclusion, the proposed SwarmFusionDNN ensemble approach for brain tumor detection presents substantial advancements in both accuracy and performance when compared to individual models and other ensemble techniques. By incorporating ConvMixer blocks into the architecture of the base models, the feature extraction capabilities are notably enhanced, allowing for the extraction of intricate spatial and channel-wise information from brain MRI scans. This integration significantly improves the accuracy and reliability of tumor classification. Rather than relying on manual experimentation, the swarm intelligence-based PSO algorithm is employed to automatically assign optimized weights to each modified base model. This optimization process results in SwarmFusionDNN achieving significantly higher classification accuracy compared to random weight assignments. Our comprehensive comparisons with other ensemble methods, including stacking, averaging, Gompertz function, and Condorcet's jury theorem, clearly highlight the effectiveness and superiority of SwarmFusionDNN in brain tumor detection. Evaluations of SwarmFusionDNN on three different brain MRI datasets demonstrated 99.66 % accuracy on the Br35H dataset, 96.14 % on the Figshare dataset, and 98.76 % on the Brain Cancer MRI dataset. The results obtained from these evaluations provide compelling evidence of the model's performance in accurately detecting and localizing brain tumors. Furthermore, the use of Grad-CAMs enhances the interpretability of our approach, visually highlighting the salient tumor regions within the brain MRI scans, which contributes to greater model transparency. In summary, this study introduces SwarmFusionDNN, a novel ensemble approach that combines the power of ConvMixer-based feature extraction and PSO-based weight optimization for improved brain tumor detection. The modifications to

pre-trained models, through the integration of ConvMixer blocks, improve their ability to extract meaningful features from brain MRI scans. The application of PSO for weight optimization further refines the ensemble's performance, yielding superior accuracy. Our approach not only outperforms other ensemble methods but also provides interpretable insights through Grad-CAM visualizations. These advancements have significant implications for the field of brain tumor detection, presenting a promising solution for accurate and reliable clinical diagnosis.

5.1. Proposed solutions for clinical deployment

- To integrate SwarmFusionDNN into clinical practice, we propose API integration via services like <https://x.ai/api>, enabling real-time predictions within hospital information systems. Deployed as a RESTful API, it can interact with EHR systems, supporting clinical decision-making.
- Interactive dashboards could be created to allow radiologists to upload MRI scans, run predictions, and visualize results, including confidence scores and Grad-CAM heatmaps. This aids in interpretation and decision-making.

5.2. Case study/simulation of clinical deployment

- In a simulated clinical environment, SwarmFusionDNN processes MRI scans from multiple patients, providing real-time predictions on an interactive dashboard. This case study would assess the model's impact on diagnostic accuracy, misdiagnosis reduction, and early detection, ultimately enhancing diagnostic speed and precision in clinical workflows.

Future work will focus on addressing class imbalance in clinical datasets through techniques like oversampling, undersampling, synthetic data generation, and class-weighted loss functions. These methods will help ensure more balanced class distributions during training and reduce bias. The potential of incorporating other architectures and advanced optimization techniques will be explored to further enhance the performance of brain tumor detection. To further improve generalization across imaging domains, future work will explore domain adaptation techniques, such as adversarial training, to reduce domain shift between different MRI acquisition protocols (e.g., T1/T2-weighted images, scanner types, and institutions). This will enhance model robustness in real-world, heterogeneous clinical environments. Furthermore, we plan to implement k-fold cross-validation to assess the model's stability and robustness across different data subsets, ensuring that the results are not overly dependent on a specific train-test split.

Declarations

Compliance with ethical standards

None.

Informed consent

None.

Data availability

The data that support the findings of this study are available from the first author upon reasonable request.

CRediT authorship contribution statement

Sohaib Asif: Data curation, Conceptualization, Writing – original draft, Validation, Software, Methodology. **Rongbiao Ying:**

Visualization, Validation, Software, Conceptualization. **Enyu Wang:** Writing – review & editing, Validation, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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